

The Limits of Consumption Taxation: A Theory of Optimal Capital Taxation*

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Abstract

In canonical models with wealth inequality, it is undesirable to tax capital: redistribution can be achieved through consumption taxation without any distortions. These models assume that consumption is observable to the government. However, in practice, business owners can disguise their consumption as a business expense to avoid or evade consumption taxes. How do such imperfections in the enforceability of consumption taxes affect optimal tax policy? We approach this question from a dynamic mechanism design perspective in general equilibrium. The key trade-off that the planner faces is that capital taxation can provide more revenue than consumption taxation, but lowers wages over time. We analytically solve for the long run state of the economy under the optimal dynamic mechanism. The optimal allocation requires sustained intertemporal distortions—as entailed, for example, by sustained capital taxation. The optimal policy can be implemented by a combination of time-varying linear taxes/subsidies on wealth, capital income, and consumption.

Keywords: capital taxation; consumption taxation; dynamic mechanism design

1 Introduction

As wealth and capital income have concentrated in the hands of a small fraction of the population (Saez and Zucman, 2016; Piketty et al., 2018), capital taxation has become increasingly salient in the public debate regarding taxation. However, as capital

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taxation causes intertemporal distortions, simple models provide arguments against using it to reduce wealth inequality (Kaldor, 1955). Rather, taxing consumption and using the revenue to lower labor income tax rates and/or fund a demogrant, would allow everyone to share in the fruits of wealth—thereby effectively expropriating initial wealth (Kaldor, 1955; Chamley, 1980; Chari et al., 2020).

In practice, however, consumption taxes are imperfect: business owners can disguise their consumption as business expenses to avoid consumption taxes. To understand why this issue arises, recall that consumption taxes are best implemented via value-added taxes (VAT), whereby businesses remit taxes on their sales to the government, but deduct taxes already paid on business inputs from the tax bill. But there is an issue of exactly which inputs should be deductible. For a VAT to be non-distortive, it is critical that all production inputs be fully deductible. Conversely, if any business expense that is not a true production input were deductible for VAT purposes, there would be a channel for tax avoidance: business owners could disguise parts of their own consumption as business expenses to avoid the VAT.¹ Similar issues arise for corporate and capital income taxes, whereby production inputs must be deductible from revenue to prevent the tax from distorting production.

Hence, the structure of consumption and corporate taxes relies on a sharp dichotomy between productive and unproductive business expenses. This distinction is absent in practice, which leads to a substantial imperfection in the consumption and corporate tax system. Indeed, Leite (2024) has provided event study evidence from an analysis of Portuguese administrative data that individuals who become owner-managers shift 31% of their household expenses to the firm—and thereby avoid consumption taxes and capital income taxes on these expenses. Closely related suggestive evidence of business owners consuming through their firms has been provided by Clotfelter (1983), Alstadsæter et al. (2014), and Bach et al. (2024).

This paper studies how the imperfection of consumption taxes affects optimal taxation when wealth is concentrated in the hands of capitalists. Our main results characterize the state of the economy in the long run under the optimal dynamic tax policy, as well as the optimal long-run distortions. Thus, our analysis characterizes the optimal balance between consumption and capital taxation in the long run.

¹For conventional sales taxes, the analogous issue is that sales tax must not be assessed on business inputs to prevent the sales tax from distorting production, which opens a channel of owners disguising their consumption as a business expense to directly avoid sales taxes.

To highlight the role of disguised consumption in disciplining the role of consumption taxation, we abstract from agency issues within businesses and model capitalists as the owner-managers of businesses that hold their wealth. We suppose that capitalists experience idiosyncratic risk, which is nonverifiable to the planner. Capitalists can spend on intermediates to increase their expected profits. However, they can also disguise their consumption as business expenses, which does not improve expected profits. We suppose that the social planner cannot distinguish productive spending from disguised consumption for the owner-manager. This imperfect observability limits the planner’s ability to redistribute via consumption taxation: business owners can avoid pure consumption taxes by disguising consumption as a business expense.²

As in Judd (1985), the planner seeks to maximize the (discounted) utility of the workers, who consume hand-to-mouth. However, the planner is constrained to provide a minimum level of utility to the capitalists—representing, e.g., political constraints.³

Rather than optimizing within a fixed set of tax instruments, we take a dynamic mechanism design approach. Thus, the model determines not only the trade-off between consumption and capital taxes, but also which tax instruments should be considered in the first place. For example, the model allows for arbitrary forms of capital taxation, including both capital income taxes and wealth taxes.

We focus on understanding the long-run state of the economy under the optimal dynamic tax policy. Define a steady state to be an optimal dynamic allocation in which the capital stock and the consumption of the workers are constant, though each capitalist may still experience idiosyncratic risk. Our main technical contribution is to prove that if the planner is sufficiently more patient than the capitalists,⁴ a steady state exists, and is in fact unique and locally saddle-path stable. Uniqueness and stability justify interpreting the steady state as a representation of the long-run state of the economy under the optimal dynamic tax policy.

Despite the complexity of the model, we can fully characterize the steady state in closed form. Three key features emerge. First, the capitalists’ total consumption as a fraction of the capital stock is proportional to how efficiently they can disguise

²Under the standard assumption that the planner could perfectly distinguish productive business spending on intermediates from unproductive business expenses that are in fact disguised owner-manager consumption (Kaldor, 1955; Chamley, 1980), the planner could implement the first best.

³As argued by Chari et al. (2020) in the context of Ramsey taxation, this promise also helps ensure time consistency in the planning problem.

⁴When the planner is not sufficiently more patient than the planner, we show that the optimal policy immiserates the worker in the long run.

consumption as a business expense. Intuitively, while the planner would prefer to allocate all consumption to workers in the long run, the capitalists must retain some consumption to deter them from disguising consumption as a business expense.

Second, the intertemporal distortion for the capitalists—i.e., their steady state *savings wedge*—is positive and takes a simple form that depends only on the discount rates and the capitalists’ elasticity of intertemporal substitution. Thus, the imperfection of consumption taxes makes long-run intertemporal distortions optimal.

Third, the capital stock is strictly below the modified golden rule level. Indeed, we show that the marginal product of capital exceeds the modified golden rule level by an amount that is increasing in both the savings wedge and the fraction of the capital stock devoted to the capitalists’ total steady state consumption. Thus, the capitalists’ consumption and the savings wedge mediate how much the moral hazard problem matters for the intertemporal aspects of the planning problem.

We also develop an implementation of the optimal policy via linear taxes/subsidies. We show that time-varying linear taxes/subsidies on wealth, (unexpected) capital income, and consumption can implement that optimal dynamic allocation. In steady state, the taxes/subsidies are constant, and we characterize them in closed form.

The optimal capital income and consumption taxes turn out to bear a simple relation that can be seen from the moral hazard problem. Specifically, the optimal taxes keep the capitalist indifferent (on the margin) between two methods to consume. First, the capitalist can consume in the open, thereby paying the consumption tax. Second, the capitalist can disguise consumption as a business expense, incurring efficiency costs but avoiding both the consumption tax and the capital income tax. To maintain this indifference, the planner faces a trade-off between taxing consumption and taxing unexpected capital income: the latter is desirable as it helps spread risk across capitalists, but limits the ability to raise revenue through the former. For example, when capitalists can disguise consumption as a business expense with relatively low costs, the optimal consumption tax is lower.

The optimal wealth tax balances two forces. First, as the planner is more patient than the capitalists, there is a mechanical motive to subsidize capital, which benefits workers in the long run by raising wages. Second, there is a motive to tax capital for redistributive purposes. How these two forces balance depends on the strength of the moral hazard problem. When it is difficult for capitalists to disguise consumption as a business expense, consumption taxes are more effective as a redistributive tool, and

the motive to subsidize capital dominates, and leads to a negative wealth tax. When it is relatively easy for capitalists to disguise consumption as a business expense, consumption taxes are less effective, so it is optimal instead to tax wealth.

To simplify our characterization of the steady state, our baseline analysis assumes that total business expenses (i.e., not merely which business expenses are productive) are not verifiable. As this assumption is not likely to hold in practice, we also consider in extension in which the planner can verify total business expenses. We characterize the steady state under the optimal dynamic tax policy and show that our qualitative conclusions regarding the distortions it entails persist. The steady state also involves an additional distortion: the planner inefficiently limits (deductible) business expenses to deter capitalists from disguising consumption as a business expense.

One of our main contributions is to develop a tractable framework for studying capital taxation in which a practical limit to consumption taxes leads to a non-trivial trade-off between capital and consumption taxes. From a methodological perspective, we obtain tractability by using stochastic calculus methods.⁵ Our formulation of the dynamic agency problem faced by the planner builds on DeMarzo and Sannikov’s (2006) continuous-time model of asset management contracts, as extended by Di Tella (2019) to settings with income effects (see also Di Tella and Sannikov (2021)).⁶ Relative to those papers, we incorporate a labor sector, which is essential to ensuring the existence of a steady state. This allows us to characterize the long run state of the economy under the optimal dynamic allocation in closed form, and to derive simple expressions for the long-run wedges and long-run implementing taxes.

Related literature. This paper relates to three strands of the literature on capital taxation. One influential strand of the literature, initiated by Chamley (1986) and Judd (1985, 1999), considers taxation of existing wealth and restricts the set of instruments available to the planner—ruling out, for example, consumption taxation, to prevent the planner from expropriating initial wealth. In such settings, the optimal long-run capital tax rate have been shown to be zero.⁷ However, for low elasticities of intertemporal substitution, the optimal long-run capital tax rate can be so high that the capital stock falls to its lowest feasible level (Straub and Werning, 2020);

⁵In the optimal tax literature, Farhi and Werning (2013) previously used stochastic calculus methods to obtain tractability in a dynamic labor income tax model.

⁶Note that Di Tella (2019) incorporated hidden trade, and Di Tella and Sannikov (2021) incorporated hidden savings, which are beyond the scope of our analysis.

⁷Similar results hold for the expected tax in models with aggregate risk (Chari et al., 1994).

for Judd’s (1985) model, moderate sustained capital taxation is optimal only in the knife-edge case in which capitalists have log utility (Lansing, 1999). By contrast, with unrestricted tax instruments and utility promises to capitalists, capital should not be taxed in steady state (Armenter, 2008; Chari et al., 2020). This paper also does not restrict the set of tax instruments, but shows that with realistic imperfections in consumption taxes, a moderate intertemporal distortion is optimal in steady state.

In this vein, two more recent papers incorporate effects of taxes on entrepreneurial activity and investment. Chari et al. (2025) develop a model in which taxes affect the investment and risk-taking decisions of entrepreneur, and show it is consistent with the dynamics of U.S. wealth inequality. They then quantitatively study the impact of taxes in dynamic general equilibrium. Amol (2024) extends Judd (1985) by allowing (as we do) capitalists to consume through their business and show that the model can match the current levels of US taxes. While we do not consider the dynamics of wealth inequality, we take a mechanism design approach that allows the planner access to all potential tax instruments and solve for the optimal dynamic tax policy.

A second strand of literature generates positive capital income taxes at optimum by supposing that agents acquire capital for precautionary savings (Aiyagari, 1995), or due to bequest motives (Piketty and Saez, 2013) or direct utility for wealth (Saez and Stantcheva, 2018). By contrast, this paper focuses on capital held by the rich, for which precautionary motives are unlikely to apply, and considers more standard preferences to isolate the interaction between consumption and capital taxation.

A third strand of literature considers the savings wedges entailed by optimal dynamic allocations when agents’ skill levels evolve stochastically over time. In such settings, when utility is separable between consumption and leisure, savings wedges are positive (Rogerson, 1985; Golosov et al., 2003). This result is driven by the fact that savings tighten future incentive constraints for labor income taxation (Golosov et al., 2006). For example, savings wedges in such models decline to 0 over the life cycle (Farhi and Werning, 2013; Stantcheva, 2017). By contrast, the moral hazard problem in this paper arises due to the riskiness of business returns, and the positivity of the savings wedge does not rely on an interaction with labor income taxation.

Paper outline. We proceed as follows. Section 2 sets up the baseline model. Section 3 characterizes the optimal dynamic allocation. Section 4 develops an implementation via linear taxes. Section 5 summarizes our solution strategy. Section 6 introduces the extension with verifiable total business expenses. Section 7 concludes.

2 Model

2.1 Environment

Time is continuous. There is a representative worker and a continuum of capitalists j , indexed by $\mathbb{J} = [0, 1]$. There is an initial capital stock K_0 ; capital depreciates at rate δ .

As in Judd (1985), the worker lives hand to mouth and inelastically supply 1 unit of labor to a competitive labor market. Letting $C_{w,t}$ denote the consumption of the representative worker at time t , the worker's utility is given by

$$\int_0^\infty e^{-\zeta t} v(C_{w,t}) dt, \quad (1)$$

where $\zeta > 0$ is a social discount rate, and $v : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ is a Bernoulli utility function.⁸ We also allow for the possibility that $\zeta = 0$, in which case the worker's utility is

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T v(C_{w,t}) dt.$$

Each capitalist j has Epstein–Zin preferences over consumption processes $C_{j,t}$, with discount rate ρ , coefficient of relative risk aversion γ , and elasticity of intertemporal substitution $\psi > 1$. For ease of exposition in the main text, we occasionally write formulae assuming CRRA preferences—i.e., $\gamma = 1/\psi$, but all results (including lemmata) are stated for the general case. In the CRRA case, capitalist j 's utility is

$$U_{j,0} \equiv \mathbb{E}_0 \int_0^\infty e^{-\rho t} C_{j,t}^{1-\frac{1}{\psi}} dt.$$

Appendix A formalizes the Epstein–Zin case, based on Duffie and Epstein (1992).

Businesses. Each capitalist is the owner-manager of a business. Without investment, if the capitalist has $K_{j,t}$ units of capital and employs $L_{j,t}$ units of labor, then their expected production at time t is $F(K_{j,t}, L_{j,t}) = K_{j,t}^\theta L_{j,t}^{1-\theta}$, where $0 < \theta < 1$ is the capital share. Writing $K_t = \int_{\mathbb{J}} K_{j,t} dj$ for the aggregate capital stock at time t , perfect competition in the labor market implies that the equilibrium wage rate at time t is $w_t = F_L(K_t, 1)$ and capitalist j 's equilibrium labor demand is $L_{j,t} = K_{j,t}/K_t$.

⁸The worker's Bernoulli utility function v is assumed to be strictly increasing, continuously differentiable, and strictly concave, and to satisfy Inada conditions.

At each point in time, a capitalist can spend on intermediates improve their expected earnings. Letting $I_{j,t}$ denote capitalist j 's intermediate spending at time t , the capitalist accrues an additional expected earnings $\alpha(I_{j,t}, K_{j,t})$, where α is strictly increasing, strictly concave, homogeneous of degree 1, and satisfies Inada conditions.

Last, each capitalist j experiences idiosyncratic risk $\sigma dZ_{j,t}$ per unit of capital, where $Z_{j,t}$ is a standard Brownian motion that is i.i.d. across j .⁹ Thus, capitalist j 's profits at time t are

$$d\Pi_{j,t} \equiv \left[\underbrace{F(K_{j,t}, L_{j,t})}_{\text{production}} - \underbrace{w_t L_{j,t}}_{\text{wages}} - \underbrace{\delta K_{j,t}}_{\text{depreciation}} + \underbrace{\alpha(I_{j,t}, K_{j,t})}_{\text{add'l earnings}} - \underbrace{I_{j,t}}_{\text{intermediates}} \right] dt + \underbrace{\sigma K_{j,t} dZ_{j,t}}_{\text{risk}}. \quad (2)$$

Disguised consumption. The capitalists can disguise consumption as business expenses, which are observably indistinguishable from intermediate spending. Letting $E_{j,t} \geq 0$ be the amount that capitalist j spends on consumption that is disguised as business expenses and $\tilde{C}_{j,t}$ their observed consumption, their (effective) consumption level is $C_{j,t} = \tilde{C}_{j,t} + \phi E_{j,t}$, where $0 < \phi < 1$ represents value of disguised consumption relative to consumption that is directly observable by the planner.

A capitalist's business expense level is $\tilde{I}_{j,t} \equiv I_{j,t} + E_{j,t}$. Thus, rather than observing the true profits $d\Pi_{j,t}$, the planner observes (and can verify) reported profits

$$d\tilde{\Pi}_{j,t} \equiv \left[\underbrace{F(K_{j,t}, L_{j,t})}_{\text{production}} - \underbrace{w_t L_{j,t}}_{\text{wages}} - \underbrace{\delta K_{j,t}}_{\text{depreciation}} + \underbrace{\alpha(I_{j,t}, K_{j,t})}_{\text{add'l earnings}} - \underbrace{\tilde{I}_{j,t}}_{\text{bus. expenses}} \right] dt + \underbrace{\sigma K_{j,t} dZ_{j,t}}_{\text{risk}}. \quad (3)$$

For tractability, our baseline model assumes that total business expenses $\tilde{I}_{j,t}$ are not verifiable. As we will show in Lemma 1, this assumption implies that, the intermediate spending level $I_{j,t}$ is efficient in each period, which allows us to solve the steady state of the economy in closed form. However, this assumption is not crucial for our main economic messages. In Section 6, we consider an extension in which total business expenses are verifiable, and derive analogues of many of our results.

Resource constraint. The resource constraint for the economy is

$$\underbrace{C_{w,t} + \int_{\mathbb{J}} \tilde{C}_{j,t} dj}_{\text{observed consumption}} + \underbrace{\int_{\mathbb{J}} \tilde{I}_{j,t} dj}_{\text{bus. expenses}} + \underbrace{\dot{K}_t + \delta K_t}_{\text{gross savings}} \leq \underbrace{\int_{\mathbb{J}} F(K_{j,t}, L_{j,t}) dj}_{\text{production}} + \underbrace{\int_{\mathbb{J}} \alpha(I_{j,t}, K_{j,t}) dj}_{\text{additional earnings}}. \quad (4)$$

⁹In our analysis, we assume the continuum version of the law of large numbers throughout.

2.2 The planning problem

We now set up the social planner's problem. An *individual plan* for capitalist j consists of trajectories $(C_{j,t}, K_{j,t})$ for their consumption and capital holdings—which can depend on the trajectory K_t of the aggregate capital stock and is adapted to (cumulative) observed idiosyncratic profits $\tilde{\Pi}_{j,t}$.

Given an individual plan, the capitalist chooses processes $E_{j,t}$ for disguised consumption and $I_{j,t}$ for intermediate spending (both adapted to the idiosyncratic shock process $Z_{j,t}$) to maximize their utility. To simplify the planning problem, we first show that the capitalist's best responses always involve efficient intermediate spending.

Lemma 1. *For any individual plan, the capitalist will (almost) always choose an efficient level of intermediate spending: i.e., all best responses almost surely satisfy*

$$I_{j,t} = \arg \max_{\tilde{I}_{j,t} \geq 0} \{ \alpha(\hat{I}_{j,t}, K_{j,t}) - \hat{I}_{j,t} \}. \quad (5)$$

To understand the intuition behind Lemma 1, suppose for sake of deriving a contradiction that a capitalist chose an inefficient investment level. The capitalist could instead choose an efficient investment level and appropriate the excess earnings as disguised consumption. Since total business expenses are not verifiable, this strategy would not change any verifiable information, so the capitalist would strictly benefit from it. We can therefore assume that $I_{j,t}$ is defined by (5) for the baseline model.

Moreover, since disguising consumption as a business expense is inefficient ($\phi < 1$) and the consumption value of disguised consumption is linear, it is without loss of generality to assume that there is no disguised consumption in equilibrium—i.e., $E_{j,t} = 0$ (DeMarzo and Sannikov, 2006; DeMarzo and Fishman, 2007). The following condition requires that $E_{j,t} = 0$ is indeed a best response for the capitalist.

Definition 1. With CRRA capitalist preferences, an individual plan $(C_{j,t}, K_{j,t})$ is *incentive-compatible* given a wage trajectory w_t if for all processes $E_{j,t}$ of disguised consumption (adapted to the idiosyncratic shock process $Z_{j,t}$), we have that

$$\mathbb{E}_0 \int_0^\infty e^{-\rho t} C_{j,t}^{1-\frac{1}{\psi}} dt \geq \hat{\mathbb{E}}_0 \int_0^\infty e^{-\rho t} (C_{j,t} + \phi E_{j,t})^{1-\frac{1}{\psi}} dt,$$

where the expectation $\mathbb{E}_0$ (resp. $\hat{\mathbb{E}}_0$) is taken assuming that observed idiosyncratic profits are given by (2) (resp. by (3) with $\tilde{I}_{j,t} = I_{j,t} + E_{j,t}$).

Appendix A extends this definition to the Epstein–Zin case.

To introduce the space of plans, we aggregate and impose the resource constraint.

Definition 2. A *plan* consists of a consumption stream $(C_{w,t})$ for the worker, and an incentive-compatible individual plan $(C_{j,t}, K_{j,t})$ for each capitalist j . Such a plan is *feasible* if the resource constraint (4) holds.

The planning problem is to choose a feasible plan to maximize the representative worker’s utility (1) for a given initial capital stock K_0 , subject to satisfying a promise to provide each capitalist j a utility level of at least $u_0 > 0$. This promise could represent, for example, a political economy constraint.¹⁰ As argued by Chari et al. (2020) in the context of Ramsey taxation, the presence of a promised utility for the capitalists help ensure that the solution to the planning problem is time-consistent.

To make the planning problem well-defined, we make one technical assumption.

Assumption 0. There exists a feasible plan with $C_{w,t} > 0$ for all t that delivers utility at least u_0 to each capitalist j .

Assumption 0 holds as long as u_0 is sufficiently small. For example, a sufficient condition is that u_0 is at most the utility level obtained in autarky if capitalists each pay for consumption, investment, and depreciation out of their own wealth.

3 The constrained optimum

In this section, we define the steady state of the constrained optimum, provide conditions under which it exists, and analytically characterize it in closed form.

3.1 Steady state

In a steady state, the capital stock and workers’ consumption are constant, but each capitalists’ consumption and capital holdings are stochastic. We therefore introduce consider the capitalists’ aggregate consumption $C_t \equiv \int_{\mathbb{J}} C_{j,t} dj$.

Definition 3. A *steady state* consists of a capital stock K^* , an aggregate promised capitalist utility U^* , a worker consumption level $C_w^* > 0$, and an aggregate capitalist consumption $C^* > 0$ such that given the initial conditions $(K_0, u_0) = (K^*, U^*)$, there exists an optimal plan $(C_{w,t}, (C_{j,t}, K_{j,t})_{j \in \mathbb{J}})$ with $(K_t, C_{w,t}, C_t) = (K^*, C_w^*, C^*)$ for all t .

¹⁰Our results would be essentially unchanged if the promised utility varied across capitalists.

Our definition of steady state requires that worker consumption and capitalist aggregate consumption each be positive in steady state, so the workers are not immiserated and the capitalists are not immiserated in aggregate.

We have three key conditions for the existence of steady state. The first condition ensures that long-run growth is infeasible. To introduce this condition, consider the maximum net excess return on capital that can be accrued via intermediate spending:

$$\alpha^* \equiv \max_{i \geq 0} \{\alpha(i, 1) - i\}. \quad (6)$$

As capitalists can achieve a net excess return of α^* , they can achieve a net return of $\alpha^* - \delta$ even under a high capital stock. To ensure that long-run growth is infeasible, we require that $\delta > \alpha^*$. In our characterizations of steady state, it will be convenient to consider an *intermediate-adjusted depreciation rate* $\delta^* \equiv \delta - \alpha^*$, that adjusts the depreciation rate for the excess return that can be accrued from intermediate spending; we then require that $\delta^* > 0$.

Assumption 1. The intermediate-adjusted depreciation rate satisfies $\delta^* > 0$.

The second assumption is needed to ensure that it is not optimal to give all the capital to a vanishingly small mass of capitalists to manage. As shown by Di Tella (2019), the key condition is that the capitalists' EIS is large enough.

Assumption 2. The elasticity of intertemporal substitution satisfies $\psi \geq 2$.

Despite the large EIS required by Assumption 2, Epstein–Zin preferences allow us to incorporate realistic levels of capitalist risk aversion in our analysis.

We also need a critical third assumption. If the capitalist is at least as patient as the planner, then the optimal allocation immiserates the worker, as in many models with infinitely repeated moral hazard (see, e.g., Atkeson and Lucas (1992)). A standard approach to restoring the existence of a steady state, based on analyses of models with a single type of agent (see, e.g., Phelan (2006) and Farhi and Werning (2007)), is to suppose that the planner is strictly more patient than the capitalist.¹¹ However, in contrast to those models, even if the planner is more patient than the capitalist in our model, the optimal allocation may still immiserate the worker.

¹¹A similar approach is usually taken in models of business cycles with financial frictions in which “experts” or “financial intermediaries” are subject to moral hazard (see, e.g., Bernanke et al. (1999)).

Proposition 1 (Worker immiseration if $\zeta \geq \rho$ or $\zeta \approx \rho$). *Suppose that Assumptions 1 and 2 hold. Fixing all parameters other than ζ , there exists $\zeta^{**} < \rho$ such that for $\zeta \geq \zeta^{**}$ and all optimal plans under planning discount rate ζ , if the quantities $K_t, C_{w,t}, C_t$ each converge to limits in $[0, \infty]$ as $t \rightarrow \infty$, then $C_{w,t} \rightarrow 0$ as $t \rightarrow \infty$.*

Intuitively, if the planner is not patient enough, it is efficient for incentive provision purposes to frontload the worker's consumption and backload the capitalists' consumption. This backloading in turn leads to the workers being immiserated in the long run. We therefore instead assume that the planner is sufficiently more patient than the worker. We formulate this condition by requiring that the ratio between the planning discount rate and the capitalists' discount rate be at most the labor share.¹²

Assumption 3. The planner's discount rate satisfies $\zeta \leq (1 - \theta)\rho$.

We can then show that there exists a unique steady state. Despite the complexity of the model, we can also completely characterize the steady state in closed form.

Theorem 1. *Under Assumptions 1–3, there exists a unique steady state, where*¹³

$$\frac{C^*}{K^*} = \phi\sigma \frac{2\rho - \zeta}{\sqrt{2(\rho - \zeta)}} \sqrt{\gamma(1 - \frac{1}{\psi})} \quad (7)$$

$$f'(K^*) = \zeta + \delta^* + \frac{1}{\frac{\zeta(1 - \frac{1}{\psi})}{2(\rho - \zeta)} + 1} \cdot \frac{C^*}{K^*} \quad (8)$$

$$\frac{C_w^*}{K^*} = (1 - \theta) \frac{f(K^*)}{K^*} + \zeta - \frac{1}{\frac{2(\rho - \zeta)}{\zeta(1 - \frac{1}{\psi})} + 1} \cdot \frac{C^*}{K^*}. \quad (9)$$

The steady state has a particularly simple characterization in the case in which the planner is very patient ($\zeta = 0$). Indeed, in this case, (7)–(9) respectively become

$$\frac{C^*}{K^*} = \phi\sigma \sqrt{2\rho\gamma(1 - \frac{1}{\psi})} \quad (10)$$

$$f'(K^*) = \delta^* + \frac{C^*}{K^*} \quad (11)$$

$$C_w^* = (1 - \theta)f(K^*). \quad (12)$$

¹²An alternative approach would be to require that the planner be strictly more patient than the worker and that the investment-adjusted depreciation rate be sufficiently large. This alternative assumption would deliver identical other results.

¹³In the steady state, we also have $U^* = (C^*)^{1 - \frac{1}{\psi}} / (2\rho - \zeta)$.

To better understand (7) and (10), note that C^*/K^* increases in the efficiency ϕ of disguising consumption as a business expense. Intuitively, the higher ϕ , the harder it is for the planner to provide incentives for the capitalists not to disguise consumption. Thus, to provide such incentives, the planner offers the capitalists a higher share of the economy, so that C^*/K^* increases with ϕ .

However, this incentive provision matters for the steady state capital stock. Equations (8) and (11) are a version of the (modified) golden rule, adjusted for the fact that capitalists also consume in the steady state. Indeed, in the first best ($\phi = 0$), the steady state level of capital would satisfy the modified golden rule $f'(K^*) = \zeta + \delta^*$. In the case of $\zeta = 0$ (see (11)), the ratio $\frac{C^*}{K^*}$ quantifies how much the steady state rate of return deviates from the golden rule level; in the case of $\zeta > 0$ (see (8)), the deviation of the steady state rate of return from the modified golden rule level also depends on preference parameters. In either case, as a higher ϕ means devoting a larger fraction C^*/K^* of resources to capitalist consumption, it also induces a lower steady-state level of capital K^* . In turn, a lower level of capital generates lower wages and (according to (9) and (12)) lower consumption for the workers, which is the cost of redistributing through capital taxation.

3.2 The savings wedge

To characterize the distortions entailed by optimal plans more sharply, we characterize the wedges they entail for capitalists' savings. Formally, capitalist j 's *savings wedge* $\omega_{j,t}^s$ at time t is defined as the difference between the risk-free rate entailed by capitalist j 's consumption process and the intertemporal marginal rate of substitution (IMRS) of the social planner, which, under CRRA preferences, is given by

$$\omega_{j,t}^s dt \equiv \underbrace{\left(\zeta dt - \frac{d(v'(C_{w,t}))}{v'(C_{w,t})} \right)}_{\text{planner's IMRS}} - \underbrace{\left(\rho dt - \mathbb{E}_t \frac{d(C_{j,t}^{-\frac{1}{\psi}})}{C_{j,t}^{-\frac{1}{\psi}}} \right)}_{\text{capitalists' risk-free rate}}.$$

Appendix A extends this definition to the Epstein–Zin case.

In optimum, the capitalists share the same savings wedge at each point in time.

Proposition 2. *Under Assumptions 0–2, for all optimal plans, at almost all times t , almost all capitalists almost surely share the same savings wedge ω_t^s .*

Intuitively, Proposition 2 follows from the homotheticity of the model. While the savings wedge is not simply characterized outside the steady state, it turns out that in the steady state, the savings wedge is positive and has a very simple formula.

Theorem 2. *Under Assumptions 1–3, under any optimal plan corresponding to a steady state, almost everywhere, we have*

$$\omega_{j,t}^s = \frac{2(\rho - \zeta)}{\psi - 1} \equiv \omega_{ss}^s. \quad (13)$$

Theorem 2 states that the steady state savings wedge is proportional to the difference between the private and social discount rates; the constant of proportionality depends on the capitalists' elasticity of intertemporal substitution. In particular, the optimal dynamic tax policy features sustained intertemporal distortions in steady state. These distortions are not mechanically driven by a motivation to increase savings when the social planner is more patient than the capitalist: indeed, the steady state savings wedge is higher the more patient the planner is relative to the capitalist.

3.3 Stability

While our steady state analysis is tractable, one may be concerned about whether they characterize the long run state of the economy under the optimal dynamic tax policy. This concern may be particularly relevant to capital taxation given the existence of an unstable steady state in the Chamley (1986) and Judd (1985) models for low elasticities of intertemporal substitution, as shown by Straub and Werning (2020).

Thus, to support our analytical results about steady states, we provide a stability analysis of the steady state. We show that if the planner is sufficiently patient (i.e., if ζ is small), then the steady state (K^*, U^*, C_w^*, C^*) is locally saddle-path stable in that the optimal plan $(C_{w,t}, (C_{j,t}, K_{j,t})_{j \in \mathbb{J}})$ converges to (K^*, U^*, C_w^*, C^*) , provided that the initial level of capital K_0 and the promised capitalist utility u_0 are sufficiently close to their respective steady state levels K^* and U^* . To that end, we first recall formalize local saddle-path stability when the steady state can feature idiosyncratic shocks.

Definition 4. A steady state (K^*, U^*, C_w^*, C^*) is *locally saddle-path stable* if there exists a neighborhood $\mathcal{N}$ of (K^*, U^*) such that if the $(K_0, u_0) \in \mathcal{N}$, then given any optimal plan $(C_{w,t}, (C_{j,t}, K_{j,t})_{j \in \mathbb{J}})$, we have $\lim_{t \rightarrow \infty} (K_t, C_{w,t}, C_t) = (K^*, C_w^*, C^*)$.

We then establish local saddle-path stability for a sufficiently patient planner.

Theorem 3. *Under Assumptions 1 and 2, if the planner's discount rate ζ is sufficiently small, then the steady state is locally saddle-path stable.*

In light of Theorem 3, we can interpret our results about the steady state as characterizing the long run state of the economy under the optimal dynamic tax policy, provided that the initial conditions are not too far from steady state.

4 Implementation

We next develop of implementation of the optimal dynamic allocation via linear taxes/subsidies on the capitalists. We show that the constrained optimum can be implemented by a combination of time-varying linear taxes/subsidies on wealth, capital income, and consumption. We also derive the steady-state taxes in closed form.

Formally, the tax system we consider is as follows:

- a tax on observed consumption $\tilde{C}_{j,t}$ with rate τ_t^c
- a tax on unexpected reported profits $d\tilde{\Pi}_{j,t} - \mathbb{E}_t[d\Pi_{j,t}]$ with rate τ_t^k
- a wealth tax on capital stocks $K_{j,t}$ with rate $\tau_t^w dt$.

We use a tax on unexpected reported profits, rather than all reported profits, to disentangle and isolate the wealth taxation and risk-sharing roles of capital taxation.¹⁴

Given these taxes, capitalists are allowed to freely save, invest, and consume. Thus, the law of motion of the capitalist j 's wealth is

$$\begin{aligned} dK_{j,t} &= d\Pi_{j,t} - \tau_t^k (d\tilde{\Pi}_{j,t} - \mathbb{E}_t[d\Pi_{j,t}]) - \tau_t^w K_{j,t} dt - (1 + \tau_t^c) \tilde{C}_{j,t} dt - E_{j,t} dt \\ &= d\tilde{\Pi}_{j,t} - \tau_t^k (d\tilde{\Pi}_{j,t} - \mathbb{E}_t[d\Pi_{j,t}]) - \tau_t^w K_{j,t} dt - (1 + \tau_t^c) \tilde{C}_{j,t} dt. \end{aligned} \quad (14)$$

Given a path of taxes $(\tau_t^c, \tau_t^k, \tau_t^w)$ and an initial capital stock $K_{j,0} = K_0$ each capitalist j chooses $(C_{j,t}, L_{j,t}, I_{j,t}, E_{j,t})$ to maximize their lifetime utility subject to the law of motion for capital and the usual no Ponzi condition.

We first establish that we can implement the optimal dynamic allocation in general equilibrium with only these three taxes.

¹⁴Working with a tax on all reported profits instead would only require correspondingly adjusting the wealth tax by the capital income tax due on expected reported profits.

Proposition 3. *Under Assumptions 0–2, there exist a unique trajectory of taxes $(\tau_t^c, \tau_t^k, \tau_t^w)_t$ that implements any constrained optimum plan.*

To understand the logic behind Proposition 3, we first summarize how to solve for the capital income tax τ_t^k . Intuitively, the capital income tax is set so that the capitalist receives the right amount of skin-in-the-game: $\tau_t^k = 1$ would imply perfect risk-sharing of the idiosyncratic risk $dZ_{j,t}$, while $\tau_t^k = 0$ would imply the capitalist j bears all the risk of the technology associated with capital $K_{j,t}$. More precisely, since the capital holdings $K_{j,t}$ under the optimal allocation follows an Itô process adapted to $Z_{j,t}$, it can be written as

$$\frac{dK_{j,t}}{K_{j,t}} = \mu_{K_j} dt + \sigma_{K_j} dZ_{j,t},$$

where σ_{K_j} is the idiosyncratic proportional volatility of capital. Then, the capital income tax τ_t^k must be set so that the idiosyncratic proportional volatility of capital reaches its level under the optimal dynamic allocation. Substituting the formulae for true and reported profits $d\Pi_{j,t}$, $\tilde{d\Pi}_{j,t}$ from (2) and (3) into (14) yields that

$$\begin{aligned} dK_{j,t} = & [F(K_{j,t}, L_{j,t}) - w_t L_{j,t} - \delta K_{j,t} + \alpha(I_{j,t}, K_{j,t}) - I_{j,t}] dt \\ & + (1 - \tau_t^k) \sigma_{K_j} K_{j,t} dZ_{j,t} - \tau_t^w K_{j,t} - (1 + \tau_t^c) C_{j,t} dt \\ & + [\phi(1 + \tau_t^c) - (1 - \tau_t^k)] E_{j,t} dt, \end{aligned} \quad (15)$$

so the proportional volatility of capital is $(1 - \tau_t^k)\sigma$. It follows that we must have

$$(1 - \tau_t^k)\sigma = \sigma_{K_j}. \quad (16)$$

Next, we describe how the presence of the possibility of consumption disguise through business expenses $(E_{j,t})$ imposes a tight link between the consumption and capital income tax rates, which helps us pins down the implementing consumption tax rate. Since $\phi < 1$, the optimal allocation implies that the capitalists do not disguise consumption through business expenses ($E_{j,t} = 0$). An inspection of the last term of (15) implies that the taxes must then satisfy $(1 - \tau_t^k) \geq \phi(1 + \tau_t^c)$. A strict inequality would require the capitalist to strictly prefer consuming observably to engaging in disguised consumption, which we show to be inconsistent with the optimal allocation, as it would involve providing excessive risk to the capitalist.

We can therefore establish the following result.

Proposition 4 (Limits of Consumption Taxation). *Under Assumptions 0–2, the implementing taxes satisfy*

$$(1 - \tau_t^k) = \phi(1 + \tau_t^c). \quad (17)$$

Equation (17) captures a key force of our model: to make the agent indifferent between disguising consumption as a business expense and consuming, the marginal capital income retention ratio must equal the product of the gross consumption tax with the efficiency of disguising consumption (ϕ). Indeed, obtaining an effective consumption of ϕ through disguised consumption requires incurring business expenses of 1, which costs $1 - \tau_t^k$ after capital income taxes, while obtaining an effective consumption of ϕ by observable consumption costs $\phi(1 + \tau_t^c)$. This equation also highlights the trade-off that the planner faces between risk-sharing and raising revenue. Taxing unexpected capital income improves risk-sharing, and therefore allows the planner to provide less consumption to capitalists, but limits the planner's ability to raise revenue through consumption taxation—a limit to consumption taxation.

Equation (17) also allows us to calculate consumption taxes from capital income taxes. Given capital income and consumption taxes, imposing the law of motion of the capitalists' wealth on the optimal dynamic allocation pins down the wealth tax.

4.1 Steady State Taxes

While the implementing taxes are not analytically tractable along the transition path to steady state, we can characterize the the steady state taxes in closed form.

Theorem 4. *Under Assumptions 1–3, any optimal plan corresponding to the steady state can be implemented with constant linear taxes $(\tau_{SS}^c, \tau_{SS}^k, \tau_{SS}^w)$ given by*

$$1 + \tau_{SS}^c = \frac{1}{\phi\sigma} \sqrt{\frac{2(\rho - \zeta)}{\gamma \left(1 - \frac{1}{\psi}\right)}} \quad (18)$$

$$1 - \tau_{SS}^k = \frac{1}{\sigma} \sqrt{\frac{2(\rho - \zeta)}{\gamma \left(1 - \frac{1}{\psi}\right)}} \quad (19)$$

$$\tau_{SS}^w = -2(\rho - \zeta) + \phi\sigma \sqrt{\frac{\gamma \left(1 - \frac{1}{\psi}\right)}{2(\rho - \zeta)}} \cdot \frac{2\rho - \zeta}{\frac{\zeta(1 - \frac{1}{\psi})}{2(\rho - \zeta)} + 1}. \quad (20)$$

As with the steady state allocation, the steady state taxes have a simpler characterization in the case in which the planner is very patient ($\zeta = 0$). Indeed, in this case, equations (18)–(20) become

$$\begin{aligned} 1 + \tau_{SS}^c &= \frac{1}{\phi\sigma} \sqrt{\frac{2\rho}{\gamma \left(1 - \frac{1}{\psi}\right)}} \\ 1 - \tau_{SS}^k &= \frac{1}{\sigma} \sqrt{\frac{2\rho}{\gamma \left(1 - \frac{1}{\psi}\right)}} \\ \tau_{SS}^w &= -2\rho + \phi\sigma \sqrt{2\rho\gamma \left(1 - \frac{1}{\psi}\right)}. \end{aligned}$$

To better understand the steady-state taxes, we consider the comparative statics of the taxes in the value ϕ of disguised consumption. The consumption tax rate is inverse to ϕ and the wealth tax rate is linear in ϕ . Indeed, with very low ϕ , the moral hazard problem is not significant, and consumption taxation is a powerful redistributive tool. The planner can then set a very high consumption tax rate without incentivizing capitalists to consume through their business, as in Kaldor’s (1955) ideal consumption tax. As a result, the planner, being more patient than the capitalists, subsidizes wealth to encourage capital accumulation and higher long run wages.

Thus, as ϕ increases, consumption taxes become less effective. The steady state capital income tax rate is unchanged, so the consumption tax rate must decrease. The wealth subsidy then decreases—and eventually becomes a wealth tax. Intuitively, the ineffectiveness of the consumption taxes due to the possibility of disguising consumption as a business expense forces the planner to turn to a wealth tax to effect redistribution, despite the intertemporal distortions that it entails.

Next, we consider comparative statics in the profit volatility σ . When profits are more volatile, the moral hazard problem is also worse, as it is harder for the planner to make inferences about the level of disguised consumption from reported profits. Analogously to raising ϕ , raising σ lowers the consumption tax and raises the wealth tax. However, raising σ also makes it more desirable for the planner to share risk across capitalists, and therefore raises the consumption tax rate as well. Similar comparative statics apply in the capitalists’ risk aversion γ , as more risk aversion makes risk-sharing more valuable and therefore worsens the moral hazard problem.

4.2 Numerical Example

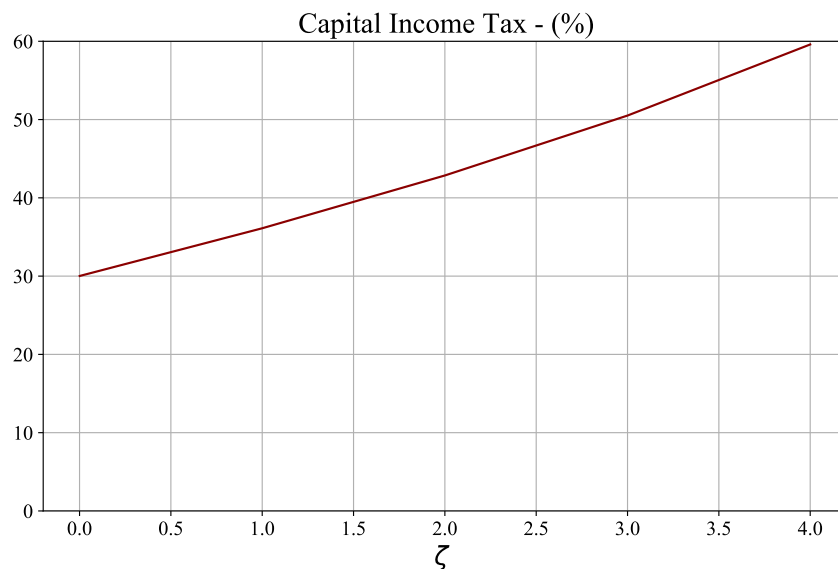
We next quantitatively illustrate the steady state taxes. To do so, we use standard value for parameters $(\rho, \gamma, \psi, \sigma)$, reported in Table 1. We then consider ranges or the value ϕ of disguising consumption and the planner's discount rate ζ , which are important and harder to pin down.

We consider a range of planner's discount rates $\zeta \in (0, 4\%)$. This implies a steady state capital income tax in the range $\tau_{SS}^k \in (30\%, 60\%)$, as depicted in Figure 1. The upper bound for ζ is the maximum value for which Assumption 3 is satisfied when the capitalists' discount rate is $\rho = 6\%$ and the capital share in production is $\theta = 1/3$. For a baseline case of $\zeta = 2\%$, we obtain a capital income tax rate of $\tau_{SS}^k = 42.85\%$.

parameter	symbol	value	source
capitalist discount rate	ρ	6%	standard
capitalist risk aversion	γ	3	standard
capitalist EIS	ψ	3	standard
volatility of profits	σ	35%	Moskowitz and Vissing-Jørgensen (2002)

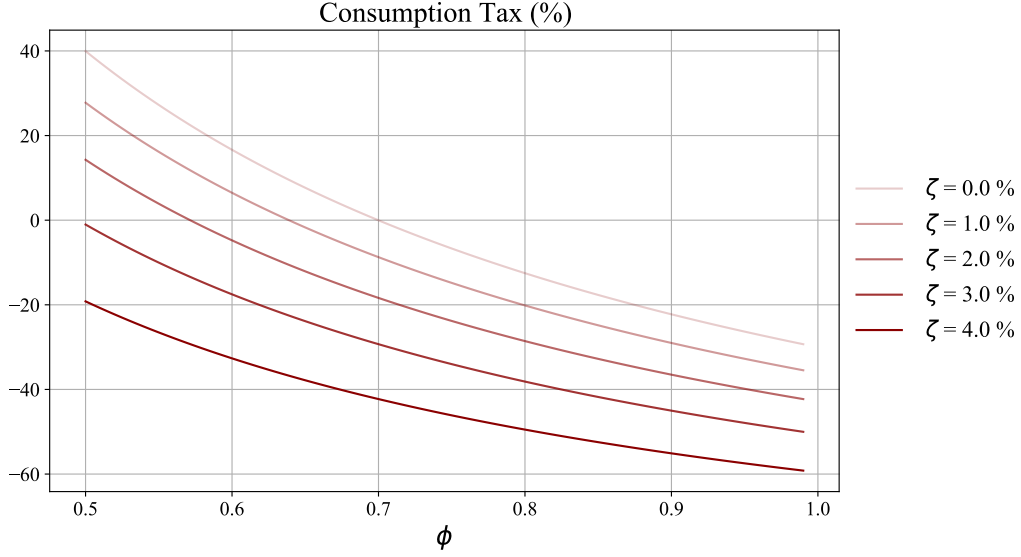
Table 1: Parameters for numerical example.

Figure 1



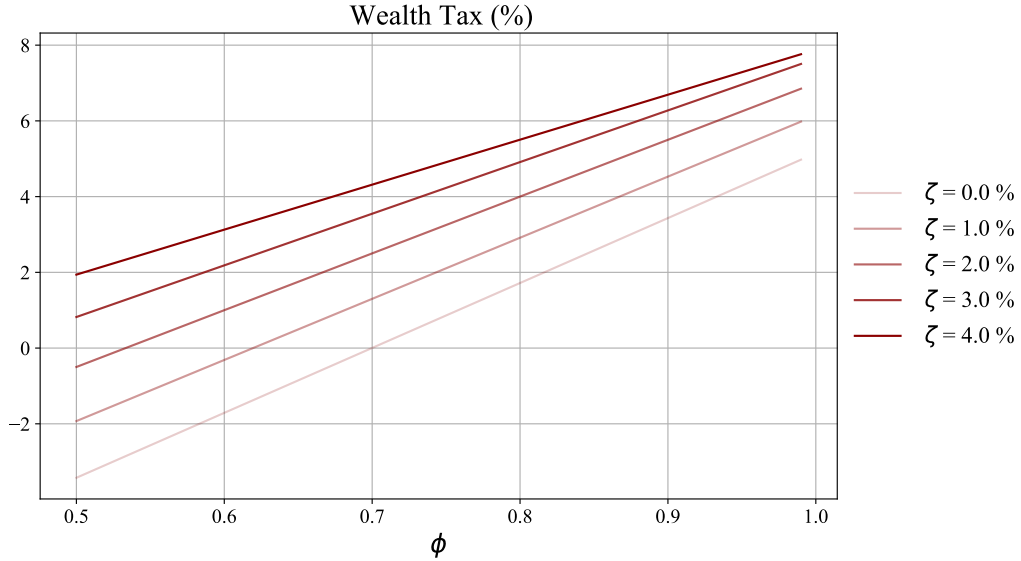
Notes: the y-axis is the capital income tax τ_{SS}^k , reported in %, whose formula is given in equation (19). The x-axis is the planner's intertemporal discount rate ζ , reported in %. For the remaining parameters, we assume the values reported in Table 1.

Figure 2



Notes: the y-axis is the consumption tax τ_{SS}^c , reported in %, whose formula is given in equation (18). The x-axis is the value ϕ of disguised consumption. The different curves consider different intertemporal discount rate ζ for the planner, with more transparent curves associated with lower values of ζ . For the remaining parameters, we assume the values reported in Table 1.

Figure 3



Notes: the y-axis is the wealth tax τ_{SS}^w , reported in %, whose formula is given in equation (20). The x-axis is the value ϕ of disguised consumption. The different curves consider different intertemporal discount rate ζ for the planner, with more transparent curves associated with lower values of ζ . For the remaining parameters, we assume the values reported in Table 1.

We consider a range for the value of disguised consumption $\phi \in (0.5, 1)$. We then use equations (18) and (20) to plot the corresponding steady-state consumption tax τ_{SS}^c and wealth tax τ_{SS}^w , depicted in Figures 2 and 3. The optimal levels of these taxes are quite sensitive to ϕ , as that parameter controls the strength of the moral hazard problem. In a baseline case where $\zeta = 2\%$ and $\phi = 0.5$, the consumption tax is $\tau_{SS}^c = 14.28\%$ and the wealth tax is $\tau_{SS}^w = -0.5\%$. An increase of ϕ from 0.5 to 0.6 would decrease the consumption tax τ_{SS}^c from 14.28% to -4.76% and increase the wealth tax τ_{SS}^w from -0.5% to 1% . As ϕ increases further, the consumption tax rate continues to fall and the wealth tax rate continues to increase.

With a very patient planner ($\zeta = 0$), the consumption tax are higher: they range from from 40% (with $\phi = 0.5$) to -30% (with $\phi = 1$). Wealth taxes are in turn lower, which reflects that a patient planner is more willing to defer consumption to accumulate capital and raise wages in the long run.

5 Analysis

In this technical section, we summarize our overall strategy to solve the model. We show that the homotheticity of preferences and technologies can be used to the planner's problem as a control problem. We use the HJB equation for this control problem to analytically characterize the steady state, and provide conditions for its existence and stability. Detailed arguments are in Appendix B.

5.1 Incentive compatibility

To begin, we need to characterize the incentive compatible individual plans. It will be useful to consider the the evolution of capitalists' *continuation utilities*, which under CRRA preferences (i.e, $\gamma = 1/\psi$), satisfies

$$U_{j,t} = \mathbb{E}_t \int_t^\infty e^{-\rho(s-t)} C_{j,s}^{1-\frac{1}{\psi}} ds.$$

The following standard lemma (see, e.g., Propositions 1 and 2 in Sannikov (2008) or Lemmata 1 and 2 in Di Tella (2019)) characterizes the law of motion of continuation utility, and expresses incentive compatibility as a condition on the law of motion.

Lemma 2. *For any individual plan $(C_{j,t}, K_{j,t})$ for a capitalist j for which $U_{j,0}$ is well-defined, capitalist j 's continuation utility process $U_{j,t}$ satisfies*

$$dU_{j,t} = \left(\rho U_{j,t} - C_{j,t}^{1-\frac{1}{\psi}} + \frac{\left(\gamma - \frac{1}{\psi}\right) \sigma_{j,t}^2}{2 \left(1 - \frac{1}{\psi}\right) U_{j,t}} \right) dt + \sigma_{j,t} dZ_{j,t} \quad (21)$$

for some stochastic process $\sigma_{j,t}$. Such an individual plan is incentive compatible if and only if the inequality

$$\sigma_{j,t} \geq \frac{\left(1 - \frac{1}{\psi}\right) \phi \sigma K_{j,t}}{C_{j,t}^{\frac{1}{\psi}}}, \quad (22)$$

holds almost surely for almost all t .

Intuitively, the inequality (22) corresponds to a skin-in-the-game constraint (Sannikov, 2008). This constraint is tighter the higher the value of disguised consumption—i.e., the worse the moral hazard problem is.

5.2 Reduction to a control problem

We now reduce the planner's problem to a control problem by exploiting the homotheticity of preferences and technologies. The state variables for this control problem are the capital stock K_t , and a particular aggregate of capitalists' continuation utilities¹⁵ given by

$$X_t = \int_{\mathbb{J}} X_{j,t} dj,$$

where $X_{j,t} \equiv U_{j,t}^{\frac{1}{1-\frac{1}{\psi}}}$ is capitalist j 's continuation utility measured in consumption units. The control variables are aggregate capitalist consumption $C_t = \int_{\mathbb{J}} C_{j,t} dj$ and worker consumption $C_{w,t}$.

The reduction to a control problem of this type relies on Assumption 2, which requires that the capitalists' EIS ψ be at least 2. This condition is needed to ensure that the planner does not desire to implement plans in which a vanishing mass of the capitalists hold all the capital (Di Tella, 2019). The key consequence of this assumption is the following lemma, which states roughly that in optimal plans, individual plans must be homogeneous in $X_{j,t}$.

¹⁵The use of continuation utilities as a state variable is motivated by Spear and Srivastava (1987).

Proposition 5. *Under Assumptions 2, for each optimal plan, we have that*

$$(C_{j,t}, K_{j,t}) = \left(\frac{X_{j,t}}{X_t} \right) (C_t, K_t) \quad (23)$$

almost surely for almost all capitalists j .

Proposition 5 leads to a reduction of the planning problem to a control problem with state variables K_t and X_t . Restricting to plans that satisfy the homogeneity condition (23), the resource constraint (4) can be written as

$$\dot{K}_t \leq f(K_t) - \delta^* K_t - C_t - C_{w,t}, \quad (24)$$

where recall that $\delta^* = \delta - (\alpha(\iota, 1) - 1)$ and $f(K) \equiv F(K, 1)$.

Moreover, combining Lemma 2 and Itô's Lemma also shows that the inequality

$$\dot{X}_t \geq \frac{\rho X_t - C_t^{1-\frac{1}{\psi}} X_t^{\frac{1}{\psi}}}{1 - \frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K_t^2}{2 C_t^{\frac{2}{\psi}} X_t^{1-\frac{2}{\psi}}} \quad (25)$$

must hold. We therefore consider a deterministic control problem for each K_0, X_0 :

$$V(K_0, X_0) \equiv \max_{(C_t, C_{w,t}, \dot{K}_t, \dot{X}_t)} \int_0^\infty e^{-\zeta t} v(C_{w,t}) dt$$

subject to (24), (25), and $\lim_{t \rightarrow \infty} e^{-\rho t} X_t = 0$. (26)

Building on Lemma 2 and Proposition 5, the following lemma relates the social planner's problem to the control problem (26)—showing how to construct optimal plans from solutions to (26).

Lemma 3. *Fix an initial capital stock K_0 . Under Assumptions 1 and 2, for any solution $(C_t, C_{w,t}, K_t, X_t)$ to the control problem (26) for $X_0 = u_0^{\frac{1}{1-\frac{1}{\psi}}}$, there exist continuation utility trajectories $X_{j,t}$ such that the trajectory $C_{w,t}$ and the individual plans defined by (23) comprise an optimal plan.*

Proposition 5 and Lemma 3 show that the control problem (26) entirely captures the planning problem. The key to our analysis is then to exploit the tractability of the control problem (26).

We can then obtain a Hamilton–Jacobi–Bellman (HJB) equation: $V(K, X)$ solves

$$\zeta V(K, X) = \max_{C, C_w} \left\{ v(C_w) + V_K (f(K) - \delta^* K - C - C_w) + V_X \left(\frac{\rho X - C^{1-\frac{1}{\psi}} X^{\frac{1}{\psi}}}{1 - \frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K^2}{2C^{\frac{2}{\psi}} X^{1-\frac{2}{\psi}}} \right) \right\}. \quad (27)$$

Under Assumption 2, we can show that (27) is a concave maximization problem. We proceed by using (27) to characterize the steady state and establish its local stability.

5.3 Steady state and stability

A steady state for the HJB problem (27) is defined by level of capital and aggregate value of promised utility to the capitalists (K^*, X^*) such that for the optimal control trajectories $C_t = C(K_t, X_t)$ and $C_{w,t} = C_w(K_t, X_t)$ that solves (27), we have that $\dot{K}_t = \dot{X}_t = 0$ for almost all t . Under Assumptions 1–3, we use the first-order conditions with respect to (C_w, C) , together with the Envelope Theorem to establish the steady state described in Theorem 1, where the associated steady-state aggregate value of capitalists’ promised utility is $X^* = C^*/(2\rho - \zeta)^{\frac{1}{1-\frac{1}{\psi}}}$.

In light of Lemma 3, taking the initial capital stock to be $K_0 = K^*$ and the promised utility to capitalists to be $u_0 = (X^*)^{1-\frac{1}{\psi}}$, there are optimal plans corresponding to the steady state. In such plans, each capitalists’ individual plan features stochastic consumption and capital holdings. However, the aggregate consumption of the capitalists and the capital stock are constant, as is the worker’s consumption.

The assumption that the planner be sufficiently patient (Assumption 3) is critical to the existence of a steady state. Indeed, we show that if the planner’s discount rate ζ is too close to or above the capitalist’s discount rate ρ , and the states and controls of (26) each converge in optimum, then the worker must be immiserated (Proposition 1).

To establish local saddle-path stability (Theorem 3), we use the HJB equation (27) to derive a non-linear dynamical system for $(C_t, C_{w,t}, K_t, X_t)$. We then log-linearize this system around the steady state and use a standard linear dynamical system argument to establish stability. Specifically, the log-linearized system is of the form $\dot{y}_t = M_{4 \times 4} y_t$ where $y_t \equiv \left(\frac{C_t - C^*}{C^*}, \frac{C_{w,t} - C_w^*}{C_w^*}, \frac{K_t - K^*}{K^*}, \frac{X_t - X^*}{X^*} \right)$. A sufficient condition for local saddle-path stability is that $M_{4 \times 4}$ have two eigenvalues with positive real parts and two eigenvalues with negative real parts. We analytically verify that this condition holds if the planner’s discount rate ζ is sufficiently small.

6 Extension: Verifiable total business expenses

In this section, we introduce an extension of our baseline model in which total business expenses $\tilde{I}_{j,t}$ are verifiable to the social planner. While natural in practice, verifiability of total business expenses makes the model less tractable and prevents us from explicitly solving for a steady state in closed form. Nevertheless, we are able to implicitly characterize the steady state and the distortions it entails. Our results are similar to the baseline model, but involve an distortion along an additional margin: to deter disguising consumption as a business expense, the planner limits (deductible) business expenses, so spending on intermediates becomes inefficiently low.

We now set up the planning problem with verifiable total business expenses. An *individual plan with verifiable total business expenses* for capitalist j consists of trajectories $(C_{j,t}, \tilde{I}_{j,t}, K_{j,t})$ for their consumption, business expenses, and capital holdings—which can depend on the trajectory K_t of the aggregate capital stock and is adapted to (cumulative) observed idiosyncratic profits $\tilde{\Pi}_{j,t}$.

Given an individual plan with total verifiable business expenses, a capitalist's choice of processes $E_{j,t}$ for disguised consumption and $I_{j,t}$ for intermediate spending is constrained to satisfy $E_{j,t} + I_{j,t} = \tilde{I}_{j,t}$. Due to this constraint, capitalists no longer choose efficient levels of intermediate spending in general (i.e., Lemma 1 fails). We therefore need to explicitly track capitalists' intermediate spending in our analysis.

It remains without loss of generality to assume that there is no disguised consumption in equilibrium—i.e., $E_{j,t} = 0$. The following condition requires that $E_{j,t} = 0$ is indeed a best response for the capitalists. Unlike in the baseline model, due to the verifiability of total business expenses $\tilde{I}_{j,t} = I_{j,t} + E_{j,t}$, engaging in disguised consumption requires lowering intermediate spending $I_{j,t}$.

Definition 5. With CRRA capitalist preferences, an individual plan with verifiable total business expenses $(C_{j,t}, \tilde{I}_{j,t}, K_{j,t})$, is *incentive-compatible* given a wage trajectory w_t if for all processes $E_{j,t} \leq \tilde{I}_{j,t}$ of disguised consumption (adapted to the idiosyncratic shock process $Z_{j,t}$), we have that

$$\mathbb{E}_0 \int_0^\infty e^{-\rho t} C_{j,t}^{1-\frac{1}{\psi}} dt \geq \hat{\mathbb{E}}_0 \int_0^\infty e^{-\rho t} (C_{j,t} + \phi E_{j,t})^{1-\frac{1}{\psi}} dt,$$

where the expectation $\mathbb{E}_0$ (resp. $\hat{\mathbb{E}}_0$) is taken assuming that observed idiosyncratic profits are given by (2) with $I_{j,t} = \tilde{I}_{j,t}$ (resp. by (3) with $I_{j,t} = \tilde{I}_{j,t} + E_{j,t}$).

Appendix A extends this definition to the Epstein–Zin case.

We can then define plans and steady states with verifiable total business expenses by aggregating similarly to the baseline model (see Appendix C for details).

The key additional force present in this extension is that the planner can affect the marginal additional return of intermediate spending by adjusting the ratio $\tilde{I}_{j,t}/K_{j,t}$. In particular, by lowering $\tilde{I}_{j,t}/K_{j,t}$, the planner can increase the capitalists' marginal value of intermediate spending, and hence raise their cost of substituting from productive spending to disguised consumption. The shape of the additional return function governs the planner's efficiency costs of doing so, and therefore plays a central role in the extension. To ensure that the planner does not want to give all the capital to a vanishingly small mass of capitalists to manage, we impose a condition on $\alpha(I_{j,t}, K_{j,t})$.

Assumption 4. We have that $\alpha(I_{j,t}, K_{j,t}) = \alpha_0 I_{j,t}^{1-\eta} K_{j,t}^\eta$, where $\eta \geq \frac{1}{2}$.

Assumption 4 can be relaxed to avoid the Cobb–Douglas functional form assumption; we impose it for simplicity. However, the bound on η implies a lower bound of 2 on the magnitude of the elasticity of the marginal additional returns from intermediate spending, which plays an essential role in ensuring the existence of a steady state similar to that of Assumption 2.

Imposing Assumptions 1–4, we can then prove the existence and uniqueness of a steady state with verifiable total business expenses, and analytically characterize it.

Theorem 5. *Under Assumptions 1–4, there exists a unique steady state with verifiable total business expenses $(K^*, U^*, C_w^*, C^*, \tilde{I}^*)$. This steady state satisfies¹⁶*

$$\frac{C^*}{K^*} = \phi\sigma \frac{2\rho - \zeta}{\sqrt{2(\rho - \zeta)}} \sqrt{\gamma(1 - \frac{1}{\psi})} \left/ \frac{\partial \alpha(I, K^*)}{\partial I} \right|_{I=\tilde{I}^*} \quad (28)$$

$$f'(K^*) = \zeta + \delta - \frac{\alpha(\tilde{I}^*, K^*) - \tilde{I}^*}{K^*} + \frac{1}{\frac{\zeta(1-\frac{1}{\psi})}{2(\rho-\zeta)} + 1} \cdot \frac{C^*}{K^*} \quad (29)$$

$$\left. \frac{\partial \alpha(I, K^*)}{\partial I} \right|_{I=\tilde{I}^*} = 1 + \frac{1}{\frac{\zeta(1-\frac{1}{\psi})}{2(\rho-\zeta)} + 1} \cdot \frac{C^*}{\tilde{I}^*} \cdot \eta \quad (30)$$

$$\frac{C_w^*}{K^*} = (1 - \theta) \frac{f(K^*)}{K^*} + \zeta - \frac{1}{\frac{2(\rho-\zeta)}{\zeta(1-\frac{1}{\psi})} + 1} \cdot \frac{C^*}{K^*}. \quad (31)$$

Terms colored in blue highlight differences from the baseline model (Theorem 1).

¹⁶In the steady state, as with the baseline model, we also have $U^* = (C^*)^{1-\frac{1}{\psi}} / (2\rho - \zeta)$.

As in the baseline model, we exposit Theorem 5 by focusing on the case in which the planner is very patient ($\zeta = 0$). In this case, (28)–(31) become

$$\frac{C^*}{K^*} = \phi\sigma\sqrt{2\rho\gamma(1 - \frac{1}{\psi})} \left/ \frac{\partial\alpha(I, K^*)}{\partial I} \right|_{I=\tilde{I}^*} \quad (32)$$

$$f'(K^*) = \delta - \frac{\alpha(\tilde{I}^*, K^*) - \tilde{I}^*}{K^*} + \frac{C^*}{K^*} \quad (33)$$

$$\left. \frac{\partial\alpha(I, K^*)}{\partial I} \right|_{I=\tilde{I}^*} = 1 + \frac{C^*}{\tilde{I}^*} \cdot \eta \quad (34)$$

$$C_w^* = (1 - \theta)f(K^*). \quad (35)$$

(32) is similar to the analogous equation (10) from the baseline model, except for a division by the steady-state marginal value of intermediate spending. Intuitively, raising the equilibrium marginal value of intermediate spending by restricting total business expenses raises the capitalists' marginal cost of disguising consumption as a business expense. This limit slackens incentive constraints, and hence allows the planner to implement allocations with less consumption for the capitalist. However, such a limit comes at the cost of distorting businesses' intermediate spending, which, in turn lowers the effective return on capital, and hence lowers the steady state capital stock and wages. This second effect can be seen by comparing (33) with (11): distorting businesses' intermediate spending inefficiently lowers the additional return from intermediate spending ($\frac{\alpha(\tilde{I}^*, K^*) - \tilde{I}^*}{K^*} < \alpha^*$).

The key new object to be characterized is therefore the steady-state degree of distortion of intermediate spending under verifiable total business expenses. (34) characterizes this distortion implicitly: difference between the steady-state marginal value of additional intermediate spending, relative to its efficient value of 1, follows an inverse elasticity rule. The relevant elasticity is the elasticity of productive intermediate spending—which, under Assumption 4, is $\frac{1}{\eta}$. The inverse elasticity η is scaled by $\frac{C^*}{K^*}$, which quantifies the relative sizes of the benefit from distorting spending (i.e., lowering C^*) and the cost of doing so (i.e., lowering K^*).

Hence, the steady state is determined by same key forces as in the baseline model, with additional effects from limiting total business expenses. In terms of an implementation via taxes, a limit on business expenses corresponds to a limit on consumption and corporate income tax deductions for business expenses.

Note that the characterization of $\tilde{I}^*$ in (34) involves $\tilde{I}^*$ on the right-hand side:

while (34) (when combined with (32)) gives a unique solution for $\frac{\tilde{I}^*}{K^*}$, the solution is not in closed form. Thus, unlike in the baseline model, we do not have an explicit characterization of steady state, even for a very patient planner ($\zeta = 0$).

Our key result characterizing the savings wedge carries over unchanged. Thus, the role for sustained intertemporal distortions under imperfect consumption tax enforceability is unchanged by the verifiability of total business expenses.

Theorem 6. *With verifiable total business expenses, under Assumptions 1–4, under any optimal plan corresponding to a steady state, almost everywhere, we have*

$$\omega_{j,t}^s = \frac{2(\rho - \zeta)}{\psi - 1}.$$

7 Conclusion

A classical argument against capital taxation is that redistribution can be done more efficiently via consumption taxes than via capital taxes. This argument assumes that consumption is perfectly observable, which is not true in practice: business owners can disguise consumption taxes as a business expense—a channel that has been shown to be relevant in practice. This paper studies the optimal form of taxation when the tax authority cannot distinguish business expenses that are disguised consumption from business expenses that are productive spending on intermediates by the business.

Our main analytical results characterize the steady state of the economy under the optimal dynamic tax policy, and show that the optimal allocation can be implemented by linear taxes/subsidies on capital income, consumption, and wealth. Our framework highlights that the ineffectiveness of consumption taxation creates a role for capital taxation. Thus, the planner faces a trade-off between raising revenue from capital taxation and lowering the capital stock (and hence wages). The optimal policy balances this trade-off, and involves both consumption and capital taxes. When total business expenses are verifiable, it is optimal to inefficiently limit them to deter capitalists from disguising consumption as a business expense.

Our analysis provides a step toward a more comprehensive investigation of optimal capital taxation. Despite its complexity, our model can be studied tractably by using stochastic calculus methods. An interesting direction for future work would be to study other trade-offs in the design of capital taxation by building on our framework.

A Epstein–Zin preferences

This appendix develops our model for the case in which the capitalists have Epstein–Zin preferences with discount rate ρ , relative risk aversion γ , and elasticity of intertemporal substitution ψ . Formally, given an effective consumption process $C_{j,t}$, capitalist j 's utility process $U_{j,t}$ is defined by the (stochastic) integral equation

$$U_{j,t} = \mathbb{E}_t \left[\int_t^\infty \left(C_{j,s}^{1-\frac{1}{\psi}} - \rho U_{j,s} - \frac{\gamma - \frac{1}{\psi}}{2(1 - \frac{1}{\psi})U_{j,s}} \frac{d}{ds} [U_j]_s \right) ds \right], \quad (\text{A.1})$$

where $[U_j]_t$ denotes the quadratic variation process of $U_{j,t}$ (Duffie and Epstein, 1992).¹⁷ The utility of capitalist j at time 0 is then $U_{j,0}$.

An individual plan $(C_{j,t}, K_{j,t})$ is *incentive-compatible* given a wage trajectory w_t if for all processes $E_{j,t}$ of disguised consumption (adapted to the idiosyncratic shock process $Z_{j,t}$), we have that $U_{j,0} \geq \hat{U}_{j,0}$, where $U_{j,t}$ solves (A.1) and $\hat{U}_{j,t}$ solves

$$\hat{U}_{j,t} = \hat{\mathbb{E}}_t \left[\int_t^\infty \left((C_{j,s} + E_{j,s})^{1-\frac{1}{\psi}} - \rho \hat{U}_{j,s} - \frac{\gamma - \frac{1}{\psi}}{2(1 - \frac{1}{\psi})\hat{U}_{j,s}} \frac{d}{ds} [\hat{U}_j]_s \right) ds \right]. \quad (\text{A.2})$$

Here, the expectations $\mathbb{E}_t$ (resp. $\hat{\mathbb{E}}_t$) are taken assuming that observed idiosyncratic profits are given by (2) (resp. by (3) with $\tilde{I}_{j,t} = I_{j,t} + E_{j,t}$).

Similarly, an individual plan with verifiable total business expenses $(C_{j,t}, \tilde{I}_{j,t}, K_{j,t})$, is *incentive-compatible* given a wage trajectory w_t if for all processes $E_{j,t} \leq \tilde{I}_{j,t}$ of disguised consumption, we have that $U_{j,0} \geq \hat{U}_{j,0}$, where $U_{j,t}$ solves (A.1) and $\hat{U}_{j,t}$ solves (A.2). Here, the expectations $\mathbb{E}_t$ (resp. $\hat{\mathbb{E}}_t$) are taken assuming that observed idiosyncratic profits are given by (2) with $I_{j,t} = \tilde{I}_{j,t}$ (resp. by (3) with $I_{j,t} = \tilde{I}_{j,t} + E_{j,t}$).

The definition of the savings wedge needs to be extended as the capitalists' risk-free rate is different with Epstein–Zin preferences. For individual plans that are Markov and homogenous of degree 1 in $X_{j,t}$, capitalist j 's *savings wedge at time t* is defined as

$$\omega_{j,t}^s dt \equiv \underbrace{\left(\zeta dt - \frac{d(v'(C_{w,t}))}{v'(C_{w,t})} \right)}_{\text{planner's IMRS}} - \underbrace{\left(\rho dt - \mathbb{E}_t \frac{d(C_{j,t}^{-\frac{1}{\psi}})}{C_{j,t}^{-\frac{1}{\psi}}} - \left(\gamma - \frac{1}{\psi} \right) \frac{1 + \frac{1}{\psi}}{2} \frac{\text{Var}_t dC_{j,t}}{C_{j,t}^2} \right)}_{\text{capitalists' risk-free rate}}. \quad (\text{A.3})$$

¹⁷Theorem C1 in Duffie and Epstein (1992) guarantees the existence and uniqueness of an integrable semi-martingale solution $U_{j,t}$ to this integral equation provided that $\mathbb{E}_0 \int_0^\infty e^{-\rho t} C_{j,t}^2 < \infty$.

B Proofs for the baseline model

Proof of Lemma 1.

Proof. Consider any individual plan. Suppose, for the sake of contradiction, that for some capitalist j , its choice for intermediate spending $I_{j,t}$ does not satisfy equation (5) with positive probability. Consider the following deviation for the capitalist: change investment to the efficient level defined in (5), which we denote $I_{j,t}^{new}$ and change the disguised consumption so that reported profits $d\tilde{\Pi}_{j,t}$ are unchanged. The difference between the new and old disguised consumption is

$$E_{j,t}^{new} - E_{j,t}^{old} = \alpha(I_{j,t}^{new}, K_{j,t}) - I_{j,t} - (\alpha(I_{j,t}, K_{j,t}) - I_{j,t}) > 0,$$

where the inequality follows from (5). It then follows that capitalist j is strictly better with the proposed deviation, a contradiction. $\square$

Proof of Theorem 1.

Proof. The proof strategy is as follows. First, we use the first-order conditions from (27) and the Envelope Theorem to derive the (candidate) steady-state equilibrium. Then, we manipulate the equilibrium conditions to derive (7)-(9). To show existence, we use Assumptions 1-3 to show that the candidate steady-state equilibrium satisfies $C_w^* > 0$, which concludes the proof.

It is convenient to make the following change of variable. Letting $c_t \equiv C_t/X_t$, equation (27) simplifies to

$$\zeta V(K, X) = \max_{c, C_w} \left\{ \begin{aligned} &v(C_w) + V_K (f(K) - \delta^* K - cX - C_w) \\ &+ V_X \left(\frac{\rho X - c^{1-\frac{1}{\psi}} X}{1 - \frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K^2}{2c^{\frac{2}{\psi}} X} \right) \end{aligned} \right\} \quad (\text{B.1})$$

The first order conditions with respect to C_w and c are

$$v'(C_w) = V_K, \quad (\text{B.2})$$

and

$$V_K X = -V_X \left(c^{-\frac{1}{\psi}} X + \frac{\gamma \phi^2 \sigma^2 K^2}{\psi c^{1+\frac{2}{\psi}} X} \right) \quad (\text{B.3})$$

Applying the Envelope Theorem in (B.1) at the steady state level yields

$$\zeta V_K = V_K(f'(K) - \delta^*) + V_X \frac{\gamma \phi^2 \sigma^2 K}{c^{\frac{2}{\psi}} X}, \quad (\text{B.4})$$

and

$$\zeta V_X = -V_K c + V_X \left(\frac{\rho - c^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} - \frac{\gamma \phi^2 \sigma^2 K^2}{2c^{\frac{2}{\psi}} X^2} \right) \quad (\text{B.5})$$

Since in the steady state $\dot{K} = \dot{X} = 0$, we also have the steady state relations

$$f(K) = \delta^* K - cX - C_w, \quad (\text{B.6})$$

and

$$\frac{\rho - c^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} = -\frac{\gamma \phi^2 \sigma^2 K^2}{2c^{\frac{2}{\psi}} X^2}. \quad (\text{B.7})$$

Using (B.3) to substitute V_K in (B.5) yields

$$\zeta = c^{1-\frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K^2}{\psi c^{\frac{2}{\psi}} X^2} + \frac{\rho - c^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} - \frac{\gamma \phi^2 \sigma^2 K^2}{2c^{\frac{2}{\psi}} X^2}.$$

Using (B.7), we obtain that the steady state level for the normalized capitalist consumption follows

$$c^* = (2\rho - \zeta)^{\frac{1}{1-\frac{1}{\psi}}}. \quad (\text{B.8})$$

Next, we can isolate X/K in (B.7) and substitute (B.8) to find that

$$\frac{X^*}{K^*} = \sqrt{\frac{\gamma(1 - \frac{1}{\psi})}{2(\rho - \zeta)}} \frac{\phi \sigma}{(2\rho - \zeta)^{\frac{1}{\psi-1}}} \quad (\text{B.9})$$

Then, isolating V_X in (B.3) and substituting it into (B.4) yields

$$\zeta = f'(K) - \delta^* - \frac{c}{c^{1-\frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K^2}{\psi c^{\frac{2}{\psi}} X^2}} \frac{\gamma \phi^2 \sigma^2 K}{c^{\frac{2}{\psi}} X}.$$

Substituting for c from (B.8) and X/K from (B.9), we obtain that the steady state

level of capital is

$$f'(K^*) = \zeta + \delta^* + \frac{2\rho - \zeta}{\zeta + 2\frac{\rho - \zeta}{1 - \frac{1}{\psi}}} \sqrt{\frac{2\gamma(\rho - \zeta)}{1 - \frac{1}{\psi}}} \phi\sigma \quad (\text{B.10})$$

We then substitute (B.10) into (B.9) to obtain X^* . Finally, using (B.6), we obtain that $C_{w,SS}$ follows

$$C_{w,SS} = f(K^*) - \delta^* K^* - c^* X^*. \quad (\text{B.11})$$

Manipulating (B.8)-(B.11) and using that $f'(K) = \theta f(K)/K$, we obtain (7)-(9) in Theorem 1. We conclude by arguing that $C_w^* > 0$. From (B.11) and $c^* = C^*/X^*$, it is enough to show that

$$\frac{f(K^*)}{K^*} > \delta^* - \frac{C^*}{K^*},$$

since $f'(K) = \theta f(K)/K$, it is enough to show that

$$f'(K^*) > \theta \delta^* - \theta \frac{C^*}{K^*}.$$

Using (7) and (B.10), the previous equation is equivalent to

$$\zeta + \delta^* + \frac{2\rho - \zeta}{\zeta + 2\frac{\rho - \zeta}{1 - \frac{1}{\psi}}} \sqrt{\frac{2\gamma(\rho - \zeta)}{1 - \frac{1}{\psi}}} \phi\sigma > \theta \delta^* - \theta \phi\sigma \frac{2\rho - \zeta}{\sqrt{2(\rho - \zeta)}} \sqrt{\gamma(1 - \frac{1}{\psi})},$$

which, after some manipulation, yields

$$\frac{2\frac{\rho - \zeta}{1 - \frac{1}{\psi}}(1 - \theta) - \theta\zeta}{\zeta + 2\frac{\rho - \zeta}{1 - \frac{1}{\psi}}} (2\rho - \zeta) \phi\sigma \sqrt{\frac{\gamma(1 - \frac{1}{\psi})}{\rho - \zeta}} + \zeta + (1 - \theta)\delta^* > 0. \quad (\text{B.12})$$

A sufficient condition for that inequality to hold is that the numerator in the first term is positive, so that

$$\theta\zeta < 2\frac{\rho - \zeta}{1 - \frac{1}{\psi}}(1 - \theta).$$

The right side of that inequality is a decreasing function of ψ , so it is enough to show that it holds under $\psi = \infty$, so we want to show that

$$\theta\zeta < 2(\rho - \zeta)(1 - \theta),$$

which holds by Assumption 3. □

Proof of Proposition 1

Proof. Denote by $X^* = \lim X_t \in [0, \infty]$. We break the proof into two cases.

Consider first the case $X^* = 0$. The proof uses the fact that the production function is Cobb-Douglas to bound capital from above, which in turn bounds the limiting capitalists' and workers' consumption. From optimal incentive provision, any path X_t that converges to $X^* = 0$ is associated with $K^* = 0$, which in turn implies $C^* = C_w^* = 0$.

To see why, given K_0 , any feasible capital path K_t is bounded from above, use that $\dot{K}_t \leq f(K_t) - \delta K^*$ from (24) and $f(K) = K^\theta$, for $\theta < 1$ to construct such bound as $\bar{K} = \max\{K_0, \delta^{-\frac{1}{1-\theta}}\}$. Since $K^* \leq \bar{K}$, then using (24) again with $\lim_t \dot{K}_t = 0$, we conclude that $C^* \leq f(\bar{K}) - \delta \bar{K}$. Then from (25), together with $\dot{X} = 0$ and (C^*, K^*) being both bounded from above, we conclude that $K^* = 0$. It then follows that $C^* = C_w^* = 0$.

Consider now the case $X^* > 0$. The proof follows by verifying that the inequality in (B.12), which is the necessary and sufficient condition for $C_w^* > 0$, is not satisfied if $\zeta \geq \rho$ or $\zeta \geq \zeta^{**}$, for some $\zeta^{**} < \rho$ sufficiently large. If the inequality in (B.12) does not hold, and the quantities $K_t, C_{w,t}, C_t$ each converge to some limit, it must be that the constraint $C_{w,t} \geq 0$ binds and the workers are immiserated in such limit.

To see why the inequality in (B.12) does not hold for $\zeta \geq \rho$, note that the first term in (B.12) is not defined for $\zeta > \rho$. For ζ approaching ρ from bellow, the first fraction in the first term in (B.12) converges to

$$\lim_{\zeta \uparrow \rho} \frac{2 \frac{\rho - \zeta}{1 - \frac{1}{\psi}} (1 - \theta) - \theta \zeta}{\zeta + 2 \frac{\rho - \zeta}{1 - \frac{1}{\psi}}} = -\theta < 0,$$

so the first term in (B.12) converges to $-\infty$ and that inequality does not hold. The same limit argument implies that there exists $\zeta^{**} < \rho$ such that (B.12) does not hold for any $\zeta \geq \zeta^{**}$, which concludes the proof. □

Proof of Lemma 2

Proof. First, we establish that (21) holds, where $U_{j,t}$ is defined by equation (A.1).

Note that $U_{j,t}$ is an Ito process, with associated drift $\mu_{j,t}$ and volatility $\sigma_{j,t}$, so that

$$dU_{j,t} = \mu_t dt + \sigma_t dZ_{j,t}.$$

Thus, to show that (21), it is enough to show that $\mu_{j,t}$ satisfies

$$\mu_t = \rho U_{j,t} - C_{j,s}^{1-\frac{1}{\psi}} + \frac{\left(\gamma - \frac{1}{\psi}\right)}{2\left(1 - \frac{1}{\psi}\right) U_{j,s}} \frac{d}{dt}[U_j]_t, \quad (\text{B.13})$$

where $\frac{d}{dt}[U_j]_t = \sigma_{j,t}^2$. To see why this is the case, note that we $U_{j,t}$ satisfies

$$U_{j,T} - U_{j,t} = \int_t^T \mu_s ds + \int_t^T \sigma_s dZ_{j,s}.$$

Taking expectation conditional on t , the limit as $T \rightarrow \infty$, with the transversality condition that $\lim_{T \rightarrow \infty} \mathbb{E}_t U_{j,T} = 0$, we obtain that

$$U_{j,t} = \mathbb{E}_t \int_t^\infty -\mu_s ds.$$

Comparing this equation with (A.1), since it must hold almost surely, we obtain (B.13).

Next, we show that an individual plan $(C_{j,t}, K_{j,t})$ is incentive compatible if and only if the inequality (22) holds. The proof follows closely Proposition 2 in Sannikov (2008). Consider an individual plan $(C_{j,t}, K_{j,t})$, measurable with respect to $\tilde{\Pi}_{j,t}$ and under which there is no disguise. Suppose that from time 0 to t , the capitalist j contemplates an alternative path with positive disguise, so that $E_{j,t} \geq 0$. Under this alternative plan, its value function is $\tilde{U}_{j,0} = \mathbb{E}_0 G_t$, where

$$G_t \equiv \int_0^t \left((C_{j,s} + \phi E_{j,s})^{1-\frac{1}{\psi}} - \rho \tilde{U}_{j,s} - \frac{\left(\gamma - \frac{1}{\psi}\right)}{2\left(1 - \frac{1}{\psi}\right) \tilde{U}_{j,s}} \frac{d}{ds}[\tilde{U}_j]_s \right) ds + e^{-\rho t} U_{j,t}.$$

The original plan is incentive compatible if and only if G_t is a martingale under the original plan and a super-martingale for any alternative choice $E_{j,t} \geq 0$. Thus, under $E_{j,t} = 0$, from the Martingale Representation Theorem, there exist a process Δ_t ,

adapted to $Z_{j,t}$, under which we have that

$$\begin{aligned}
dG_t &= \left(C_{j,t}^{1-\frac{1}{\psi}} - \rho U_{j,t} - \frac{\left(\gamma - \frac{1}{\psi}\right)}{2\left(1 - \frac{1}{\psi}\right) U_{j,t}} \frac{d}{ds} [U_j]_s \right) dt + d(e^{-\rho t} U_{j,t}) \\
&= \Delta_t \sigma K_{j,t} dZ_{j,t} \\
&= \Delta_t [d\Pi_{j,t} - (F(K_{j,t}, L_{j,t}) - w_t L_{j,t} - \delta K_{j,t} + \alpha(I_{j,t}, K_{j,t}) - I_{j,t})],
\end{aligned}$$

where the last line follows from (2), where $L_{j,t} = K_{j,t}/K_t$ and $I_{j,t}$ satisfies (5). It then follows that

$$\begin{aligned}
d(e^{-\rho t} U_{j,t}) &= - \left(C_{j,t}^{1-\frac{1}{\psi}} - \rho U_{j,t} - \frac{\left(\gamma - \frac{1}{\psi}\right)}{2\left(1 - \frac{1}{\psi}\right) U_{j,t}} \frac{d}{ds} [U_j]_s \right) dt \\
&\quad + \Delta_t [d\Pi_{j,t} - (F(K_{j,t}, L_{j,t}) - w_t L_{j,t} - \delta K_{j,t} + \alpha(I_{j,t}, K_{j,t}) - I_{j,t})]
\end{aligned}$$

Under the alternative path $E_{j,t} \geq 0$, we have instead that G_t satisfies

$$\begin{aligned}
dG_t &= \left((C_{j,t} + E_{j,t})^{1-\frac{1}{\psi}} - \rho \tilde{U}_{j,t} - \frac{\left(\gamma - \frac{1}{\psi}\right)}{2\left(1 - \frac{1}{\psi}\right) U_{j,t}} \frac{d}{ds} [\tilde{U}_j]_s \right) dt + e^{-\rho t} d(e^{-\rho t} U_{j,t}) \\
&= \left((C_{j,t} + E_{j,t})^{1-\frac{1}{\psi}} - \rho U_{j,t} - \frac{\left(\gamma - \frac{1}{\psi}\right)}{2\left(1 - \frac{1}{\psi}\right) U_{j,t}} \frac{d}{ds} [U_j]_s \right) dt \\
&\quad - \left(C_{j,t}^{1-\frac{1}{\psi}} - \rho \tilde{U}_{j,t} - \frac{\left(\gamma - \frac{1}{\psi}\right)}{2\left(1 - \frac{1}{\psi}\right) U_{j,t}} \frac{d}{ds} [\tilde{U}_j]_s \right) dt \\
&\quad + \Delta_t \left[d\tilde{\Pi}_{j,t} - (F(K_{j,t}, L_{j,t}) - w_t L_{j,t} - \delta K_{j,t} + \alpha(I_{j,t}, K_{j,t}) - I_{j,t}) \right].
\end{aligned}$$

Where the second line uses the definition of $d(e^{-\rho t} U_{j,t})$ derived above, replacing $d\Pi_{j,t}$ for $d\tilde{\Pi}_{j,t}$, since the capitalist is now under the alternative path with $E_{j,t} \geq 0$ and the individual plans are adapted to $d\tilde{\Pi}_{j,t}$. Using the definition for $d\tilde{\Pi}_{j,t}$ in (3) yields that

dG_t follows

$$\begin{aligned} dG_t = & \left((C_{j,t} + E_{j,t})^{1-\frac{1}{\psi}} - \rho \tilde{U}_{j,t} - \frac{\left(\gamma - \frac{1}{\psi}\right)}{2\left(1 - \frac{1}{\psi}\right) U_{j,t}} \frac{d}{ds} [\tilde{U}_j]_s \right) dt \\ & - \left(C_{j,t}^{1-\frac{1}{\psi}} - \rho U_{j,t} - \frac{\left(\gamma - \frac{1}{\psi}\right)}{2\left(1 - \frac{1}{\psi}\right) U_{j,t}} \frac{d}{ds} [U_j]_s \right) dt \\ & + \Delta_t [-E_{j,t}dt + \sigma K_{j,t}dZ_{j,t}]. \end{aligned}$$

Hence, given Δ_t , the capitalist chooses $E_{j,t}$ to maximize the drift of dG_t , so that

$$E^* \in \arg \max_{E_{j,t} \geq 0} (C_{j,t} + E_{j,t})^{1-\frac{1}{\psi}} - \Delta_t E_{j,t}.$$

Hence, an allocation is incentive-compatible if and only if $E^* = 0$ is a solution to the above maximization. This occurs if and only if

$$\left(1 - \frac{1}{\psi}\right) \phi C_{j,t}^{-\frac{1}{\psi}} - \Delta_t \leq 0,$$

or equivalently, if and only if

$$\Delta_t \geq \left(1 - \frac{1}{\psi}\right) \phi C_{j,t}^{-\frac{1}{\psi}}.$$

By using that $\sigma_{j,t} = \Delta_t K_{j,t} \sigma$, we obtain that an individual plan is incentive compatible if and only if

$$\sigma_{j,t} \geq \left(1 - \frac{1}{\psi}\right) \frac{\phi \sigma K_{j,t}}{C_{j,t}^{-\frac{1}{\psi}}},$$

which coincides with (22), finishing the proof. $\square$

Proof of Proposition 2

Proof. It follows immediately from (23) that the savings wedge $\omega_{j,t}^s$, defined in (A.3), is the same across capitalists. $\square$

Proof of Theorem 2

Proof. From (A.3), recall that we can write the savings' wedge as

$$\omega_{j,t}^s dt = IMRS_t dt - r_t dt, \quad (\text{B.14})$$

where the planner's intertemporal marginal rate of substitution (IMRS) follows

$$IMRS_t dt = \zeta dt - \frac{d(v'(C_{w,t}))}{v'(C_{w,t})} = \zeta dt - \frac{v''(C_{w,t})}{v'(C_{w,t})} dC_{w,t}, \quad (\text{B.15})$$

and the capitalist's risk-free rate r_t follows

$$r_t dt = \rho dt - \mathbb{E}_t \frac{d(C_{j,t}^{-\frac{1}{\psi}})}{C_{j,t}^{-\frac{1}{\psi}}} - \left(\gamma - \frac{1}{\psi} \right) \frac{1 + \frac{1}{\psi}}{2} \frac{\text{Var}_t dC_{j,t}}{C_{j,t}^2}. \quad (\text{B.16})$$

In the steady state, (B.15) becomes $IMRS^* = \zeta$. To derive the steady state risk-free rate, r^* , we first establish that interest rate dynamics (B.16) also satisfies

$$r_t = \rho + \frac{1}{\psi} \frac{dc_t}{c_t} + \frac{1}{\psi} \frac{\dot{X}_t}{X_t} - \gamma \frac{1 + \frac{1}{\psi}}{2} \left(\frac{\phi \sigma K_t}{C_t^{\frac{1}{\psi}} X_t^{1 - \frac{1}{\psi}}} \right)^2. \quad (\text{B.17})$$

To see why, use (23) to rewrite $C_{j,t} = X_{j,t} c_t$, where $c_t \equiv C_t/X_t$, so that by Ito's Lemma we obtain that

$$\frac{\mathbb{E}_t d \left(C_{j,t}^{-\frac{1}{\psi}} \right)}{C_{j,t}^{-\frac{1}{\psi}}} = -\frac{1}{\psi} \frac{dc_t}{c_t} - \frac{1}{\psi} \frac{\mathbb{E}_t dX_{j,t}}{X_{j,t}} + \frac{1 + \frac{1}{\psi}}{2\psi} \frac{\text{Var}_t dX_{j,t}}{X_{j,t}^2}, \quad (\text{B.18})$$

and

$$\frac{\text{Var}_t dC_{j,t}}{C_{j,t}^2} = \frac{\text{Var}_t dX_{j,t}}{X_{j,t}^2}. \quad (\text{B.19})$$

Substituting (B.18) and (B.19) into (B.16) yields

$$r_t dt = \rho dt + \frac{1}{\psi} \frac{dc_t}{c_t} + \frac{1}{\psi} \frac{\mathbb{E}_t dX_{j,t}}{X_{j,t}} - \gamma \frac{1 + \frac{1}{\psi}}{2} \frac{\text{Var}_t dX_{j,t}}{X_{j,t}^2}. \quad (\text{B.20})$$

That (B.20) implies (B.17) follows from the law for $X_{j,t}$, which we derive next. To

do so, apply Ito's Lemma in $X_{j,t} = U_{j,t}^{\frac{1}{1-\frac{1}{\psi}}}$, for $U_{j,t}$ defined in (21) to obtain that

$$dX_{j,t} = \left[\frac{X_{j,t} \left(\rho - \left(\frac{C_{j,t}}{X_{j,t}} \right)^{1-\frac{1}{\psi}} \right)}{1 - \frac{1}{\psi}} + \frac{\gamma X_{j,t}^{\frac{2}{\psi}-1} \sigma_{j,t}^2}{2 \left(1 - \frac{1}{\psi} \right)^2} \right] dt + \frac{1}{1 - \frac{1}{\psi}} X_{j,t}^{\frac{1}{\psi}} \sigma_{j,t} dZ_{j,t}, \quad (\text{B.21})$$

where $\sigma_{j,t}$ satisfies (22). Substituting in $\sigma_{j,t}$ from (22), which holds with equality at optimum, yields

$$dX_{j,t} = \left[\frac{X_{j,t} \left(\rho - \left(\frac{C_{j,t}}{X_{j,t}} \right)^{1-\frac{1}{\psi}} \right)}{1 - \frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 X_{j,t}^{\frac{2}{\psi}-1} K_{j,t}^2}{2 C_{j,t}^{\frac{2}{\psi}}} \right] dt + \frac{\phi \sigma X_{j,t}^{\frac{1}{\psi}} K_{j,t}}{C_{j,t}^{\frac{1}{\psi}}} dZ_{j,t}.$$

Using (23) to substitute in for $(C_{j,t}, K_{j,t})$ in the above equation yields

$$\frac{dX_{j,t}}{X_{j,t}} = \left[\frac{\rho - \left(\frac{C_t}{X_t} \right)^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K_t^2}{2 C_t^{\frac{2}{\psi}} X_t^{2-\frac{2}{\psi}}} \right] dt + \frac{\phi \sigma K_t}{C_t^{\frac{1}{\psi}} X_t^{1-\frac{1}{\psi}}} dZ_{j,t}. \quad (\text{B.22})$$

Thus, from (B.22) we obtain that

$$\frac{\mathbb{E}_t dX_{j,t}}{X_{j,t}} = \frac{\rho - \left(\frac{C_t}{X_t} \right)^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K_t^2}{2 C_t^{\frac{2}{\psi}} X_t^{2-\frac{2}{\psi}}} = \frac{dX_t}{X_t}, \quad (\text{B.23})$$

and

$$\frac{\text{Var}_t dX_{j,t}}{X_{j,t}^2} = \left(\frac{\phi \sigma K_t}{C_t^{\frac{1}{\psi}} X_t^{1-\frac{1}{\psi}}} \right)^2 dt. \quad (\text{B.24})$$

Replacing (B.23) and (B.24) into (B.20) implies (B.17).

Evaluating (B.17) at steady state yields

$$\begin{aligned} r^* &= \rho - \gamma \frac{1 + \frac{1}{\psi}}{2} \left(\frac{\phi \sigma K^*}{(C^*)^{\frac{1}{\psi}} (X^*)^{1-\frac{1}{\psi}}} \right)^2 \\ &= \rho - \frac{1 + \frac{1}{\psi}}{1 - \frac{1}{\psi}} (\rho - \zeta), \end{aligned} \quad (\text{B.25})$$

where the second line follows from (B.8) and (B.9), for $C^* = c^* X^*$. Evaluating (B.14) at steady state and substituting in $IMRS^* = \zeta$ and r^* from (B.25) yields

$$\omega^* = \frac{2(\rho - \zeta)}{\psi - 1},$$

which coincides with (13). □

Proof of Theorem 3

Proof. The proof strategy is as follows. First, we derive the non-linear ODE system that characterizes the equilibrium dynamics for $(K_t, X_t, C_{w,t}, c_t)$. Then, by the Hartman–Grobman Theorem, the steady state is locally saddle-path stable if the (log)-linearized version of the equilibrium dynamics is saddle path-stable. To show the latter, its enough to show that log-linearized system has two positive and two negative eigenvalues, which follows from the corresponding Jacobian matrix possessing a trace equal to zero and a positive determinant.

We proceed with the following lemma, which derives the non-linear ODE system.

Lemma B.1. *Given (K_0, X_0) , the associated path for $(K_t, X_t, C_{w,t}, c_t)$ that solves (27), where $c_t \equiv C_t/X_t$ satisfies*

$$\dot{K}_t = f(K_t) - \delta^* K_t - c_t X_t - C_{w,t}, \quad (\text{B.26})$$

$$\dot{X}_t = \frac{\rho - c_t^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} X_t + \frac{\gamma \phi^2 \sigma^2 K_t^2}{2c_t^{\frac{2}{\psi}} X_t}, \quad (\text{B.27})$$

$$-\frac{v''(C_{w,t})}{v'(C_{w,t})} \dot{C}_{w,t} = f'(K_t) - \delta^* - \zeta - \frac{c_t}{c_t^{1-\frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K_t^2}{\psi c_t^{\frac{2}{\psi}} X_t^2}} \frac{\gamma \phi^2 \sigma^2 K_t}{c_t^{\frac{2}{\psi}} X_t}, \quad (\text{B.28})$$

and

$$\begin{aligned} \frac{\frac{c^{1-\frac{1}{\psi}}}{\psi} + \left(1 + \frac{2}{\psi}\right) \frac{\gamma \phi^2 \sigma^2 K^2}{\psi c^{\frac{2}{\psi}} X^2}}{c^{1-\frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K^2}{\psi c^{\frac{2}{\psi}} X^2}} \frac{\dot{c}}{c} &= -\frac{v''(C_{w,t})}{C_{w,t}} \dot{C}_{w,t} - \frac{\rho - c^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} + \left(1 - \frac{2}{\psi}\right) \frac{\gamma \phi^2 \sigma^2 K^2}{2c^{\frac{2}{\psi}} X^2} \\ &\quad + \zeta - c^{1-\frac{1}{\psi}} + 2 \frac{\frac{\gamma \phi^2 \sigma^2 K^2}{\psi c^{\frac{2}{\psi}} X^2}}{c^{1-\frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K^2}{\psi c^{\frac{2}{\psi}} X^2}} \left(\frac{\dot{K}}{K} - \frac{\dot{X}}{X} \right). \end{aligned} \quad (\text{B.29})$$

Proof. To derive (B.26) and (B.27), replace $C_t = c_t/X_t$ in (24) and (25) and use the monotonicity of the value function $V(K, X)$ on (K, X) to replace the inequalities with equalities.

To derive the law of motion for $C_{w,t}$ and c_t , we use the first order conditions of the HJB problem, (B.2) and (B.3), together with the Envelope Theorem. To that end, apply the log function on (B.2) and (B.3) and derive then with respect to time to obtain

$$\frac{v''(C_w)}{v'(C_w)} \dot{C}_{w,t} = \frac{V_{KK} \dot{K} + V_{KX} \dot{X}}{V_K}, \quad (\text{B.30})$$

and

$$\begin{aligned} \frac{V_{KK} \dot{K} + V_{KX} \dot{X}}{V_K} + \frac{\dot{X}}{X} &= \frac{V_{KX} \dot{K} + V_{XX} \dot{X}}{V_X} \\ &\quad + \frac{-\frac{1}{\psi} c^{-\frac{1}{\psi}} X \dot{c} + c^{-\frac{1}{\psi}} \dot{X} + \frac{\gamma \phi^2 \sigma^2 K^2}{\psi c^{1+\frac{2}{\psi}} X} \left[2 \frac{\dot{K}}{K} - \left(1 + \frac{2}{\psi} \right) \frac{\dot{c}}{c} - \frac{\dot{X}}{X} \right]}{c^{-\frac{1}{\psi}} X + \frac{\gamma \phi^2 \sigma^2 K^2}{\psi c^{1+\frac{2}{\psi}} X}}. \end{aligned} \quad (\text{B.31})$$

Then, apply the Envelope Theorem in (B.1) to obtain

$$\zeta V_K = V_K(f'(K) - \delta^*) + V_X \frac{\gamma \phi^2 \sigma^2 K}{c^{\frac{2}{\psi}} X} + V_{KK} \dot{K} + V_{KX} \dot{X}, \quad (\text{B.32})$$

$$\zeta V_X = -c V_K + V_X \left(\frac{\rho - c^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} - \frac{\gamma \phi^2 \sigma^2 K^2}{2 c_t^{\frac{2}{\psi}} X^2} \right) + V_{KX} \dot{K} + V_{XX} \dot{X}. \quad (\text{B.33})$$

Rearranging (B.32) we obtain

$$\begin{aligned} -\frac{V_{KK} \dot{K} + V_{KX} \dot{X}}{V_K} &= f'(K) - \delta^* - \zeta + \frac{V_X}{V_K} \frac{\gamma \phi^2 \sigma^2 K}{c^{\frac{2}{\psi}} X} \\ &= f'(K) - \delta^* - \zeta - \frac{c}{c^{1-\frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 K^2}{\psi c^{\frac{2}{\psi}} X^2}} \frac{\gamma \phi^2 \sigma^2 K}{c^{\frac{2}{\psi}} X}, \end{aligned} \quad (\text{B.34})$$

where the last equation uses (B.3) to replace for V_X/V_K . We can then obtain (B.28) by replacing (B.34) into (B.30).

Next, rearrange (B.33) to obtain

$$\begin{aligned}\frac{V_{KX}\dot{K} + V_{XX}\dot{X}}{V_X} &= -\frac{\rho - c^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} + \frac{\gamma\phi^2\sigma^2K^2}{2c_t^{\frac{2}{\psi}}X^2} + \zeta + c\frac{V_K}{V_X} \\ &= -\frac{\rho - c^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} + \left(1 - \frac{2}{\psi}\right) \frac{\gamma\phi^2\sigma^2K^2}{2c_t^{\frac{2}{\psi}}X^2} + \zeta - c^{1-\frac{1}{\psi}}, \quad (\text{B.35})\end{aligned}$$

where the second line equation uses (B.3) to replace for V_X/V_K . Replacing (B.30) and (B.35) into (B.31)

$$\begin{aligned}\frac{v''(C_{w,t})}{v'(C_{w,t})}\dot{C}_{w,t} + \frac{\dot{X}}{X} &= -\frac{\rho - c^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} + \left(1 - \frac{2}{\psi}\right) \frac{\gamma\phi^2\sigma^2K^2}{2c_t^{\frac{2}{\psi}}X^2} + \zeta - c^{1-\frac{1}{\psi}} \\ &\quad - \frac{c^{1-\frac{1}{\psi}}}{\psi} \frac{\dot{c}}{c} + c^{1-\frac{1}{\psi}} \frac{\dot{X}}{X} + \frac{\gamma\phi^2\sigma^2K^2}{\psi c^{\frac{2}{\psi}}X^2} \left[2\frac{\dot{K}}{K} - \left(1 + \frac{2}{\psi}\right) \frac{\dot{c}}{c} - \frac{\dot{X}}{X}\right] \\ &\quad + \frac{c^{1-\frac{1}{\psi}} + \frac{\gamma\phi^2\sigma^2K^2}{\psi c^{\frac{2}{\psi}}X^2}}{c^{1-\frac{1}{\psi}} + \frac{\gamma\phi^2\sigma^2K^2}{\psi c^{\frac{2}{\psi}}X^2}}.\end{aligned}$$

where we also multiplied the last term in (B.31) by c/X in the numerator and the denominator. Isolating $\dot{c}/c$ in the last equation and rearranging, we obtain (B.29). $\square$

For any variable A_t , denote $a_t = (A_t - A^*)/A^*$ the relative deviation to its steady state value A^* . In a slight abuse of notation, c_t stands for $(C_t/X_t - C^*/X^*)/(C^*/X^*)$ in the equations that follows. Log-linearizing the system (B.26-B.29) and taking the limit as $\zeta \rightarrow 0^+$ yields

$$\dot{k}_t = \frac{C^*}{K^*}k_t - \frac{C^*}{K^*}x_t - \frac{C^*}{K^*}c_t - \frac{1-\theta}{\theta} \left(\delta + \frac{C^*}{K^*}\right) c_{w,t}, \quad (\text{B.36})$$

$$\dot{x}_t = \frac{2\rho}{1 - \frac{1}{\psi}}k_t - \frac{2\rho}{1 - \frac{1}{\psi}}x_t - \frac{2\rho}{1 - \frac{1}{\psi}}c_t, \quad (\text{B.37})$$

$$\dot{c}_{w,t} = \chi \left[\left(\frac{2}{\psi} + \theta - 2 \right) \frac{C^*}{K^*} - (1 - \theta)\delta \right] k_t + \chi \frac{\psi - 2}{\psi} \frac{C^*}{K^*} x_t - \frac{\chi}{\psi^2} \frac{C^*}{K^*} c_t, \quad (\text{B.38})$$

where we write $\chi \equiv -\frac{v'(C_w^*)}{v''(C_w^*)C_w^*} > 0$, and

$$\begin{aligned} \frac{2 + \frac{1}{\psi}}{\psi} \dot{c}_t = & \left[\left(\frac{4}{\psi} - 1 \right) \left(\frac{C^*}{K^*} - \frac{2\rho}{1 - \frac{1}{\psi}} \right) - (1 - \theta) \left(\delta + \frac{C^*}{K^*} \right) \right] k_t \\ & - \left(\frac{4}{\psi} - 1 \right) \left[\frac{C^*}{K^*} - \frac{2\rho}{1 - \frac{1}{\psi}} \right] x_t \\ & - \frac{2 + \frac{1}{\psi}}{\psi} \left[\frac{C^*}{K^*} - \frac{2\rho}{1 - \frac{1}{\psi}} \right] c_t - \frac{2}{\psi} \frac{1 - \theta}{\theta} \left(\delta + \frac{C^*}{K^*} \right) c_{w,t}. \quad (\text{B.39}) \end{aligned}$$

Or equivalently, in matrix form $\dot{y}_t = A y_t$, for $y_t = \begin{pmatrix} k_t & x_t & c_t & c_{w,t} \end{pmatrix}$ and

$$A = \begin{pmatrix} s & -s & -s & -\frac{1-\theta}{\theta}(\delta + s) \\ a & -a & -a & 0 \\ \frac{\psi^2}{2\psi+1} \left[\left(\frac{4}{\psi} - 1 \right) (s - a) - (1 - \theta)(\delta + s) \right] & -\frac{\psi^2}{2\psi+1} \left(\frac{4}{\psi} - 1 \right) (s - a) & a - s & -\frac{\psi^2}{2\psi+1} \frac{2}{\psi} \frac{1-\theta}{\theta} (\delta + s) \\ \left(\left(\frac{2}{\psi} + \theta - 2 \right) s - (1 - \theta)\delta \right) \chi & \frac{\psi-2}{\psi} \chi s & -\frac{1}{\psi^2} \chi s & 0 \end{pmatrix},$$

where

$$a = \frac{2\rho}{1 - \frac{1}{\psi}} > 0 \quad \text{and} \quad s = \frac{C^*}{K^*} > 0.$$

Note that $\text{tr}(A) = 0$ and

$$\det A = \frac{a^2(\delta + s)^2(\psi - 1)^2(1 - \theta)^2\chi}{\theta(2\psi + 1)} > 0.$$

Hence, there must be exactly two positive and two negative eigenvalues for the Jacobian matrix A and the log-linearized system is saddle-path stable around zero for $\zeta = 0$. Since the eigenvalue is a continuous function of ζ , the log-linearized system is also saddle-path stable for ζ small enough. It then follows from the Hartman–Grobman Theorem that the steady state is locally saddle-path stable (in the original non-linear system). $\square$

Proof of Proposition 3

Proof. The proof strategy is as follows. We start by deriving the law for $K_{j,t}$ in the optimal allocation. We can then use such law to pin-down the capital income tax τ_t^k from (16). We then pin-down the consumption taxation τ_t^c from (17). We can

then pin-down the wealth taxation τ_t^w by ensuring that the law of motion of capital $K_{j,t}$ in the decentralized problem follows the optimal one. We conclude by showing that for such taxes, capitalist j chooses a consumption path $C_{j,t}$ that implements the constrained efficient allocation.

To derive the law for $K_{j,t}$ at optimum, use from (23) that $K_{j,t} = (X_{j,t}/X_t)K_t$ and apply Ito's Lemma to obtain that

$$\frac{dK_{j,t}}{K_{j,t}} = \frac{dX_{j,t}}{X_{j,t}} + \frac{dK_t}{K_t} - \frac{dX_t}{X_t}. \quad (\text{B.40})$$

Substituting the the laws for $(X_{j,t}, X_t, K_t)$ given by (B.22, 24, 25) into (B.40) yields

$$\frac{dK_{j,t}}{K_{j,t}} = \frac{f(K_t) - \delta^* K_t - c_t X_t - C_{w,t}}{K_t} dt + \frac{\phi \sigma K_t}{c_t^{\frac{1}{\psi}} X_t} dZ_{j,t}, \quad (\text{B.41})$$

for $c_t = C_t/X_t$.

Evaluating (16) with $\sigma_{K_j} = \frac{\phi \sigma K_t}{c_t^{\frac{1}{\psi}} X_t}$, we obtain that the capital income tax τ_t^k follows

$$1 - \tau_t^k = \frac{\phi K_t}{c_t^{\frac{1}{\psi}} X_t}. \quad (\text{B.42})$$

Next, we obtain the consumption taxation τ_t^c by substituting (B.42) into (17), which yields

$$1 + \tau_t^c = \frac{K_t}{c_t^{\frac{1}{\psi}} X_t}. \quad (\text{B.43})$$

Next, to obtain the wealth taxation τ_t^w , first substitute in (B.42) and (B.43) into (15), which yields

$$\begin{aligned} \frac{dK_{j,t}}{K_{j,t}} = & \frac{F(K_{j,t}, L_{j,t}) - w_t L_{j,t} - \delta K_{j,t} + \alpha(I_{j,t}, K_{j,t}) - I_{j,t}}{K_{j,t}} dt \\ & + \frac{\phi \sigma K_t}{c_t^{\frac{1}{\psi}} X_t} dZ_{j,t} - \tau_t^w dt - \frac{K_t}{c_t^{\frac{1}{\psi}} X_t} \frac{C_{j,t}}{K_{j,t}} dt \end{aligned} \quad (\text{B.44})$$

Recall that $w_t = F_L(K_t, 1)$, so capitalist j 's labor demand is $L_{j,t} = K_{j,t}/K_t$. Furthermore, capitalist j choose investment $I_{j,t}$ to maximize the drift of equation (B.44), so

that equation (5) holds, which together with (6) implies that (B.44) simplifies to

$$\frac{dK_{j,t}}{K_{j,t}} = [f'(K_t) - \delta^* - \tau_t^w] dt + \frac{\phi\sigma K_t}{c_t^{\frac{1}{\psi}} X_t} dZ_{j,t} - \frac{K_t}{c_t^{\frac{1}{\psi}} X_t} \frac{C_{j,t}}{K_{j,t}} dt, \quad (\text{B.45})$$

where recall that $f(K) \equiv F(K, 1)$. The wealth tax τ_t^w is such that (B.41) coincides with (B.45), so that

$$\frac{f(K_t) - \delta^* K_t - c_t X_t - C_{w,t}}{K_t} = f'(K_t) - \delta^* - \tau_t^w - \frac{K_t}{c_t^{\frac{1}{\psi}} X_t} \frac{C_{j,t}}{K_{j,t}}$$

From (23), $C_{j,t} = (K_{j,t}/K_t)C_t$ and rearranging, yields that the wealth tax follows

$$\tau_t^w = f'(K_t) - \frac{f(K_t)}{K_t} + c_t X_t + C_{w,t} - c_t^{1-\frac{1}{\psi}}. \quad (\text{B.46})$$

Finally, we argue that given the above taxes, it is optimal for capitalist j to set $C_{j,t} = (K_{j,t}/K_t)C_t$. With preferences given by (A.1) and given $K_{j,0}$, they solve

$$\max_{C_{j,t}} \mathbb{E}_0 \left[\int_0^\infty \left(C_{j,s}^{1-\frac{1}{\psi}} - \rho U_{j,s} - \frac{\gamma - \frac{1}{\psi}}{2(1 - \frac{1}{\psi}) U_{j,s}} \frac{d}{ds} [U_j]_s \right) ds \right],$$

such that $K_{j,t}$ follows (B.45), τ_t^w follows (B.46) and the determinist aggregate variables $(c_t, K_t, X_t, C_{w,t})$ follows the equilibrium path.

Since $(c_t, K_t, X_t, C_{w,t})$ are deterministic, the state variables are $K_{j,t}$ and calendar time t . Denoting $J(K_j, t)$ the value function evaluated at time t , it follows that it solves the HJB equation¹⁸

$$\begin{aligned} 0 = \max_{C_j} J_t(K_j, t) &+ \left[f'(K_t) - \delta^* - \tau_t^w - \frac{K_t}{c_t^{\frac{1}{\psi}} X_t} \frac{C_{j,t}}{K_j} \right] K_j J_{K_j}(K_j, t) + \frac{1}{2} \left(\frac{\phi\sigma K_t}{c_t^{\frac{1}{\psi}} X_t} K_{j,t} \right)^2 J_{K_j, K_j}(K_j, t) \\ &+ C_j^{1-\frac{1}{\psi}} - \rho J(K_j, t) - \frac{\gamma - \frac{1}{\psi}}{2(1 - \frac{1}{\psi}) J(K_j, t)} \left(\frac{\phi\sigma K_t}{c_t^{\frac{1}{\psi}} X_t} K_{j,t} \right)^2 J_{K_j}(K_j, t)^2. \quad (\text{B.47}) \end{aligned}$$

¹⁸See Proposition 9 in Duffie and Epstein (1992).

The first-order condition yields

$$C_j^{-\frac{1}{\psi}} = \frac{K_t}{\left(1 - \frac{1}{\psi}\right) c_t^{\frac{1}{\psi}} X_t} J_{K_j}(K_j, t). \quad (\text{B.48})$$

Lets guess and verify that

$$J(K_j, t) = \left(\frac{X_t K_j}{K_t}\right)^{1-\frac{1}{\psi}}. \quad (\text{B.49})$$

It then follows that (B.48) satisfies

$$C_j = \frac{K_j}{K_t} C_t. \quad (\text{B.50})$$

To verify our guess, substitute (B.49) and (B.50) into (B.47) to obtain that

$$\begin{aligned} 0 = \frac{d\left(\frac{X_t}{K_t}\right)^{1-\frac{1}{\psi}}}{dt} + \left[f'(K_t) - \delta^* - \tau_t^w - c_t^{1-\frac{1}{\psi}} - \frac{\gamma}{2} \left(\frac{\phi\sigma K_t}{c_t^{\frac{1}{\psi}} X_t}\right)^2 \right] \left(1 - \frac{1}{\psi}\right) \left(\frac{X_t}{K_t}\right)^{1-\frac{1}{\psi}} \\ + \left(\frac{C_t}{K_t}\right)^{1-\frac{1}{\psi}} - \rho \left(\frac{X_t}{K_t}\right)^{1-\frac{1}{\psi}}. \end{aligned}$$

Substituting in (B.46) and using (24), we obtain

$$\begin{aligned} 0 = \frac{d\left(\frac{X_t}{K_t}\right)^{1-\frac{1}{\psi}}}{dt} + \left[\frac{\dot{K}}{K} - \frac{\gamma}{2} \left(\frac{\phi\sigma K_t}{c_t^{\frac{1}{\psi}} X_t}\right)^2 \right] \left(1 - \frac{1}{\psi}\right) \left(\frac{X_t}{K_t}\right)^{1-\frac{1}{\psi}} \\ + \left(\frac{C_t}{K_t}\right)^{1-\frac{1}{\psi}} - \rho \left(\frac{X_t}{K_t}\right)^{1-\frac{1}{\psi}}. \end{aligned}$$

Dividing by $\left(\frac{X_t}{K_t}\right)^{1-\frac{1}{\psi}}$ and rearranging yields

$$\frac{d \log \left(\frac{X_t}{K_t}\right)^{1-\frac{1}{\psi}}}{dt} = \rho - \left(\frac{C_t}{X_t}\right)^{1-\frac{1}{\psi}} - \left[\frac{\dot{K}}{K} - \frac{\gamma}{2} \left(\frac{\phi\sigma K_t}{c_t^{\frac{1}{\psi}} X_t}\right)^2 \right] \left(1 - \frac{1}{\psi}\right),$$

which after rearranging yields

$$\frac{\dot{X}_t}{X_t} = \frac{\rho - \left(\frac{C_t}{X_t}\right)^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} + \frac{\gamma}{2} \left(\frac{\phi \sigma K_t}{c_t^{\frac{1}{\psi}} X_t} \right)^2,$$

which is equivalent to (25) with $c_t = C_t/X_t$, confirming our guess. $\square$

Proof of Proposition 4

Proof. To establish the result, it is enough to show that (22) holds with equality, or equivalently, that (25) holds with equality. This follows from a simple static optimization argument, formalized in the following lemma.

Lemma B.2. *In any optimum allocation $(C_{w,t}, C_t)$ that solves (26), equation (25) holds with equality almost everywhere.*

Proof. Suppose, by way of contradiction, that (25) in some non-zero measure time interval. Then, we can reduce C_t in such interval until the law binds, and increase the worker's consumption $C_{w,t}$ accordingly so that the path for K_t , and hence X_t , remain unchanged. This alternative path strictly increases the workers welfare, yielding a contradiction with the original path being optimal. $\square$

$\square$

Proof of Theorem 4.

Proof. From (B.42), (B.43) and (B.46), we have that the steady state taxes follows

$$\begin{aligned} 1 + \tau_{SS}^c &= \frac{K^*}{(c^*)^{\frac{1}{\psi}} X^*} \\ 1 - \tau_{SS}^k &= \frac{\phi K^*}{(c^*)^{\frac{1}{\psi}} X^*} \\ \tau_{SS}^w &= f'(K^*) - \delta^* - (c^*)^{1-\frac{1}{\psi}}, \end{aligned}$$

where in the last equation we also used that, in steady state, $\dot{K} = 0$.

Substituting in (B.8) and (B.9), we obtain that the above equations are equivalent to 18, 19 and 20. $\square$

Proof of Proposition 5.

Proof. Applying Ito's lemma in $X_t \equiv \int_j X_{j,t} dj$ and using (B.21) yields that

$$\dot{X}_t = \int_j \left[\frac{X_{j,t} \left(\rho - \left(\frac{C_{j,t}}{X_{j,t}} \right)^{1-\frac{1}{\psi}} \right)}{1 - \frac{1}{\psi}} + \frac{\gamma X_{j,t}^{\frac{2}{\psi}-1} \sigma_{j,t}^2}{2 \left(1 - \frac{1}{\psi} \right)^2} \right] dj.$$

where $\sigma_{j,t}$ satisfies (22). From (22), we also obtain that

$$\dot{X}_t \geq \int_j \left[\frac{X_{j,t} \left(\rho - \left(\frac{C_{j,t}}{X_{j,t}} \right)^{1-\frac{1}{\psi}} \right)}{1 - \frac{1}{\psi}} + \frac{\gamma \phi^2 \sigma^2 X_{j,t}^{\frac{2}{\psi}-1} K_{j,t}^2}{2 C_{j,t}^{\frac{2}{\psi}}} \right] dj.$$

Denote the integrand in the right side $g(X_{j,t}, K_{j,t}, C_{j,t})$. Note that, since $\psi \geq 2$, $g(\cdot)$ is an strictly convex and homogeneous of degree one function of $(X_{j,t}, C_{j,t}, K_{j,t})$. It then follows from Jensen's Inequality that

$$\dot{X}_t \geq \int_j g(X_{j,t}, K_{j,t}, C_{j,t}) dj \geq g(X_t, K_t, C_t) = \dot{X}_t,$$

where the last inequality follows from (25) binding in any optimum allocation – the proof is the same as the one used to establish Lemma B.2. It then follows that

$$\int_j g(X_{j,t}, K_{j,t}, C_{j,t}) dj = g(X_t, K_t, C_t).$$

Using that $g(\cdot)$ is homogeneous of degree one, the above equality becomes

$$\int_j X_{j,t} g \left(1, \frac{K_{j,t}}{X_{j,t}}, \frac{C_{j,t}}{X_{j,t}} \right) dj = X_t g \left(1, \frac{K_t}{X_t}, \frac{C_t}{X_t} \right). \quad (\text{B.51})$$

Since $g(1, \cdot, \cdot)$ is strictly convex (due to Assumption 2), it follows that

$$\frac{K_{j,t}}{X_{j,t}} = \frac{K_t}{X_t} \text{ and } \frac{C_{j,t}}{X_{j,t}} = \frac{C_t}{X_t},$$

almost surely which establishes (23). □

Proof of Lemma 3.

Proof. The result follows immediately from (B.22), which has an explicit solution given by

$$X_{j,t} = X_{j,0} \exp \left(\int_0^t \left[\frac{\rho - \left(\frac{C_s}{X_s} \right)^{1-\frac{1}{\psi}}}{1 - \frac{1}{\psi}} + \frac{(\gamma - 1)\phi^2 \sigma^2 K_s^2}{2C_s^{\frac{2}{\psi}} X_s^{2-\frac{2}{\psi}}} \right] dt + \int_0^t \frac{\phi \sigma K_s}{C_s^{\frac{1}{\psi}} X_s^{1-\frac{1}{\psi}}} dZ_{j,s} \right)$$

□

C Details for Section 6

In this appendix, we formalize the planning problem define steady states for the extension with verifiable total business expenses. In this appendix, we formalize the planning problem define steady states for the extension with verifiable total business expenses. The results of Section 6 are stated in terms of this formalization.

To introduce the space of plans with verifiable total business expenses, we aggregate and impose the resource constraint.

Definition C.1. A *plan with verifiable total business expenses* consists of a consumption stream $(C_{w,t})$ for the worker, and an incentive-compatible individual plan with verifiable business expenses $(C_{j,t}, \tilde{I}_{j,t}, K_{j,t})$ for each capitalist j . Such a plan is *feasible* if the resource constraint (4) holds for intermediate spending processes $I_{j,t} = \tilde{I}_{j,t}$.

The planning problem is to choose a feasible plan with verifiable business expenses to maximize the representative worker's utility (1) for a given initial capital stock K_0 , subject to satisfying a promise to provide each capitalist j utility at least $u_0 > 0$.

We can define steady states as in the baseline model, but additionally tracking aggregate spending on intermediates $I_t = \int_{\mathbb{J}} I_{j,t} dj$.

Definition C.2. A *steady state with verifiable total business expenses* consists of

- a capital stock K^* ,
- an aggregate promised capitalist utility U^* ,
- a worker consumption level $C_w^* > 0$,
- an aggregate capitalist consumption $C^* > 0$, and

- an aggregate business expense level $\tilde{I}_t$

such that given the initial conditions $(K_0, u_0) = (K^*, U^*)$, there exists an optimal plan with verifiable total business expenses $(C_{w,t}, (C_{j,t}, \tilde{I}_{j,t}, K_{j,t})_{j \in \mathbb{J}})$ such that $(K_t, C_{w,t}, C_t, \tilde{I}_t) = (K^*, C_w^*, C^*, \tilde{I}^*)$ for all t .

Proofs of Theorems 5 and 6 are to be added.

References

- Aiyagari, S. R. (1995). Optimal capital income taxation with incomplete markets, borrowing constraints, and constant discounting. *Journal of Political Economy* 103(6), 1158–1175.
- Alstadsæter, A., W. Kopczuk, and K. Telle (2014). Are closely held firms tax shelters? *Tax Policy and the Economy* 28(1), 1–32.
- Amol, A. (2024). Ramsey taxation with endogenous financial frictions. Working paper.
- Armenter, R. (2008). A note on incomplete factor taxation. *Journal of Public Economics* 92(10-11), 2275–2281.
- Atkeson, A. and R. E. Lucas (1992). On efficient distribution with private information. *Review of Economic Studies* 59(3), 427–453.
- Bach, L., A. Bozio, A. Guillouzo, C. Leroy, and C. Malgouyres (2024). Follow the money! Why dividends overreact to flat-tax reforms. Working paper.
- Bernanke, B. S., M. Gertler, and S. Gilchrist (1999). The financial accelerator in a quantitative business cycle framework. *Handbook of Macroeconomics* 1, 1341–1393.
- Chamley, C. (1980). Optimal intertemporal taxation and the public debt. Cowles Foundation Discussion Paper No. 554, Yale University.
- Chamley, C. (1986). Optimal taxation of capital income in general equilibrium with infinite lives. *Econometrica* 54(3), 607–622.
- Chari, V. V., L. J. Christiano, and P. J. Kehoe (1994). Optimal fiscal policy in a business cycle model. *Journal of Political Economy* 102(4), 617–652.
- Chari, V. V., P. J. Kehoe, P. Lopez, E. Pastorino, and S. Salgado (2025). Taxing the rich? a theory of income and wealth inequality. Working paper.

- Chari, V. V., J. P. Nicolini, and P. Teles (2020). Optimal capital taxation revisited. *Journal of Monetary Economics* 116, 147–165.
- Clotfelter, C. T. (1983). Tax-induced distortions and the business-pleasure borderline: The case of travel and entertainment. *American Economic Review* 73(5), 1053–1065.
- DeMarzo, P. M. and M. J. Fishman (2007). Optimal long-term financial contracting. *Review of Financial Studies* 20(6), 2079–2128.
- DeMarzo, P. M. and Y. Sannikov (2006). Optimal security design and dynamic capital structure in a continuous-time agency model. *Journal of Finance* 61(6), 2681–2724.
- Di Tella, S. (2019). Optimal regulation of financial intermediaries. *American Economic Review* 109(1), 271–313.
- Di Tella, S. and Y. Sannikov (2021). Optimal asset management contracts with hidden savings. *Econometrica* 89(3), 1099–1139.
- Duffie, D. and L. G. Epstein (1992). Stochastic differential utility. *Econometrica* 60(2), 353–394.
- Epstein, L. G. and S. E. Zin (1989). Substitution, risk aversion, and the temporal behavior of consumption and asset returns: A theoretical framework. *Econometrica* 57(4), 937–969.
- Farhi, E. and I. Werning (2007). Inequality and social discounting. *Journal of Political Economy* 115(3), 365–402.
- Farhi, E. and I. Werning (2013). Insurance and taxation over the life cycle. *Review of Economic Studies* 80(2), 596–635.
- Golosov, M., N. Kocherlakota, and A. Tsyvinski (2003). Optimal indirect and capital taxation. *Review of Economic Studies* 70(3), 569–587.
- Golosov, M., A. Tsyvinski, and I. Werning (2006). New dynamic public finance: A user’s guide [with comments and discussion]. *NBER Macroeconomics Annual* 21, 317–387.
- Judd, K. L. (1985). Redistributive taxation in a simple perfect foresight model. *Journal of Public Economics* 28(1), 59–83.
- Judd, K. L. (1999). Optimal taxation and spending in general competitive growth models. *Journal of Public Economics* 71(1), 1–26.
- Kaldor, N. (1955). *An Expenditure Tax*. London: Allen & Unwin.

- Lansing, K. J. (1999). Optimal redistributive capital taxation in a neoclassical growth model. *Journal of Public Economics* 73(3), 423–453.
- Leite, D. (2024). The firm as tax shelter: Micro evidence and aggregate implications of consumption through the firm. Working paper.
- Moskowitz, T. J. and A. Vissing-Jørgensen (2002). The returns to entrepreneurial investment: A private equity premium puzzle? *American Economic Review* 92(4), 745–778.
- Phelan, C. (2006). Opportunity and social mobility. *Review of Economic Studies* 73(2), 487–504.
- Piketty, T. and E. Saez (2013). A theory of optimal inheritance taxation. *Econometrica* 81(5), 1851–1886.
- Piketty, T., E. Saez, and G. Zucman (2018). Distributional national accounts: Methods and estimates for the United States. *Quarterly Journal of Economics* 133(2), 553–609.
- Rogerson, W. P. (1985). Repeated moral hazard. *Econometrica* 53(1), 69–76.
- Saez, E. and S. Stantcheva (2018). A simpler theory of optimal capital taxation. *Journal of Public Economics* 162, 120–142.
- Saez, E. and G. Zucman (2016). Wealth inequality in the United States since 1913: Evidence from capitalized income tax data. *Quarterly Journal of Economics* 131(2), 519–578.
- Sannikov, Y. (2008). A continuous-time version of the principal-agent problem. *Review of Economic Studies* 75(3), 957–984.
- Spear, S. E. and S. Srivastava (1987). On repeated moral hazard with discounting. *Review of Economic Studies* 54(4), 599–617.
- Stantcheva, S. (2017). Optimal taxation and human capital policies over the life cycle. *Journal of Political Economy* 125(6), 1931–1990.
- Straub, L. and I. Werning (2020). Positive long run capital taxation: Chamley-Judd revisited. *American Economic Review* 110(1), 86–119.